

MATA37H3 — Week 04

Integration Techniques II: Trigonometric Integrals, Partial Fractions & the Weierstrass Substitution

Recap and orientation

Last week covered substitution and integration by parts. This week finishes the integration-techniques toolkit with **trigonometric-integral identities** (for products and powers of $\sin$, $\cos$, $\sec$, $\tan$), **partial fractions** (for rational functions), and one powerful “universal” technique, the **Weierstrass substitution**, which converts *any* rational function of $\sin x$ and $\cos x$ into an ordinary rational function of t , solvable by partial fractions.

Why this matters

A $\sec^n x \tan^m x$ integral attacked with the wrong parity choice, or a partial-fraction integral where long division is forgotten because the numerator’s degree is not smaller than the denominator’s, are classic exam traps. The Weierstrass substitution is the technique to reach for when no other pattern applies: it always works, though it is not always the fastest. Units on improper integrals and series (Weeks 5-14) all assume fluency with this full toolkit.

KEY FACT Powers of $\sin x$ and $\cos x$ (OpenStax §3.2)

To integrate $\int \sin^n x \cos^m x \, dx$:

- If **either** exponent is **odd**, peel off one copy of that function, convert the remaining even power using $\sin^2 x + \cos^2 x = 1$, and substitute $u =$ (the other function). For odd m : write $\cos^m x = \cos^{m-1} x \cos x$, convert $\cos^{m-1} x$ (an even power) via $\cos^2 x = 1 - \sin^2 x$, then let $u = \sin x$.
- If **both** exponents are **even**, use the half-angle identities $\sin^2 x = \frac{1 - \cos(2x)}{2}$ and $\cos^2 x = \frac{1 + \cos(2x)}{2}$ to reduce the powers before integrating (repeat if needed).

KEY FACT Powers of $\sec x$ and $\tan x$ (OpenStax §3.2; MAT137 Unit 9)

To integrate $\int \sec^n x \tan^m x \, dx$ use $\frac{d}{dx}[\tan x] = \sec^2 x$, $\frac{d}{dx}[\sec x] = \sec x \tan x$ and $\tan^2 x + 1 = \sec^2 x$:

- If n (the power of secant) is **even**, peel off $\sec^2 x$, convert the rest of the secant power to tangents via $\sec^2 x = 1 + \tan^2 x$, and substitute $u = \tan x$.
- If m (the power of tangent) is **odd**, peel off one $\sec x \tan x$, convert the remaining even tangent power to secants via $\tan^2 x = \sec^2 x - 1$, and substitute $u = \sec x$.
- The integrals $\int \sec x \, dx = \ln |\sec x + \tan x| + C$ and $\int \csc x \, dx = -\ln |\csc x + \cot x| + C$ fit neither pattern and must be memorized (or re-derived with the trick of multiplying

by $\frac{\sec x + \tan x}{\sec x + \tan x}$).

DEFINITION Partial Fraction Decomposition (OpenStax §3.4)

A rational function $\frac{P(x)}{Q(x)}$ can be decomposed into simpler fractions **only when** $\deg P < \deg Q$. If $\deg P \geq \deg Q$, **long divide first**: $\frac{P(x)}{Q(x)} = A(x) + \frac{R(x)}{Q(x)}$ with $\deg R < \deg Q$, and only then decompose $\frac{R(x)}{Q(x)}$. The shape of the decomposition is determined by factoring $Q(x)$:

- A **distinct linear factor** $(x - a)$ contributes $\frac{A}{x - a}$.
- A **repeated linear factor** $(x - a)^k$ contributes $\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \dots + \frac{A_k}{(x - a)^k}$: one term for *every* power from 1 to k .
- An **irreducible quadratic factor** contributes $\frac{Ax + B}{x^2 + px + q}$ (a linear numerator).

Each piece then integrates with $\int \frac{du}{u} = \ln |u| + C$ (linear factors), the power rule (repeated factors), or $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$ (irreducible quadratics).

KEY FACT The Weierstrass Substitution, a Universal Rational-Trig Technique (Dummit)

For an integrand that is a rational function of $\sin x$ and $\cos x$, the substitution $t = \tan\left(\frac{x}{2}\right)$ converts the entire integrand into an ordinary rational function of t via

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2dt}{1+t^2}.$$

The resulting integral in t is always solvable by partial fractions. It is a **fallback**: it always works on rational-trig integrands, but is not always the fastest route.

Worked Examples

EXAMPLE 1

Level: Foundational. Odd power of sine

Evaluate $\int \sin^3 x \cos^2 x \, dx$.

Step 1. Identify the powers: $\sin x$ appears to the power 3 (odd) and $\cos x$ to the power 2 (even). The rule for $\int \sin^m x \cos^n x \, dx$ says that when one exponent is odd, we peel off one copy of that odd-power function.

Step 2. Peel off one factor of $\sin x$: $\sin^3 x = \sin^2 x \cdot \sin x$. The reason for keeping a single $\sin x$ is that it will become the du of a substitution.

Step 3. The remaining even power $\sin^2 x$ must be written using cosines. Use the Pythagorean identity $\sin^2 x + \cos^2 x = 1$, which gives $\sin^2 x = 1 - \cos^2 x$.

Step 4. Rewrite the integral:

$$\int \sin^3 x \cos^2 x \, dx = \int (1 - \cos^2 x) \cos^2 x \sin x \, dx.$$

Now every factor except the lone $\sin x$ is a function of $\cos x$.

Step 5. Choose the substitution $u = \cos x$. Differentiating, $du = -\sin x \, dx$, so $\sin x \, dx = -du$.

Step 6. Substitute: $\cos^2 x = u^2$ and $\sin x \, dx = -du$, so the integral becomes

$$-\int (1 - u^2) u^2 \, du.$$

Step 7. Expand the product inside the integral: $(1 - u^2)u^2 = u^2 - u^4$. So we need $-\int (u^2 - u^4) \, du$.

Step 8. Apply the power rule $\int u^k \, du = \frac{u^{k+1}}{k+1}$ to each term:

$$-\left(\frac{u^3}{3} - \frac{u^5}{5}\right) + C.$$

Step 9. Distribute the minus sign: $-\frac{u^3}{3} + \frac{u^5}{5} + C$.

Step 10. Replace u by $\cos x$ to return to the original variable: $-\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$.

Step 11. Verify by differentiating. By the chain rule, $\frac{d}{dx} \left[-\frac{\cos^3 x}{3} \right] =$
 $-\frac{3 \cos^2 x (-\sin x)}{3} = \cos^2 x \sin x$ and $\frac{d}{dx} \left[\frac{\cos^5 x}{5} \right] = \frac{5 \cos^4 x (-\sin x)}{5} = -\cos^4 x \sin x$.

Step 12. Add the two derivatives: $\cos^2 x \sin x - \cos^4 x \sin x = \cos^2 x \sin x (1 - \cos^2 x) = \cos^2 x \sin x \cdot \sin^2 x = \sin^3 x \cos^2 x$, which is the original integrand.

Answer. $-\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$

EXAMPLE 2

Level: Foundational. Both powers even: half-angle identity

Evaluate $\int \cos^2 x \, dx$.

Step 1. Identify the powers: $\cos x$ appears to the power 2 and $\sin x$ to the power 0. Both are even, so there is no odd factor to peel off and the substitution method of Example 1 is unavailable.

Step 2. Instead, lower the power with a half-angle identity. Recall the double-angle formula $\cos(2x) = 2 \cos^2 x - 1$.

Step 3. Solve that formula for $\cos^2 x$: add 1 to both sides to get $1 + \cos(2x) = 2 \cos^2 x$, then divide by 2:

$$\cos^2 x = \frac{1 + \cos(2x)}{2}.$$

Step 4. Replace the integrand using this identity:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos(2x)}{2} \, dx.$$

Step 5. Pull out the constant factor $\frac{1}{2}$ and split the sum:

$$\frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos(2x) \, dx.$$

Step 6. The first integral is $\frac{1}{2} \int 1 \, dx = \frac{x}{2}$.

Step 7. For the second integral use the substitution $w = 2x$, so $dw = 2 \, dx$ and $dx = \frac{dw}{2}$.

$$\text{Then } \int \cos(2x) \, dx = \frac{1}{2} \int \cos w \, dw = \frac{\sin w}{2} = \frac{\sin(2x)}{2}.$$

Step 8. Multiply by the outside factor $\frac{1}{2}$: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{\sin(2x)}{2} = \frac{\sin(2x)}{4}$.

Step 9. Combine the pieces and add the constant:

$$\frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

Step 10. Verify by differentiating: $\frac{d}{dx} \left[\frac{x}{2} \right] = \frac{1}{2}$ and $\frac{d}{dx} \left[\frac{\sin(2x)}{4} \right] = \frac{2 \cos(2x)}{4} = \frac{\cos(2x)}{2}$.

The sum is $\frac{1 + \cos(2x)}{2} = \cos^2 x$, as required.

Answer. $\frac{x}{2} + \frac{\sin(2x)}{4} + C$

EXAMPLE 3

Level: Intermediate. Both powers even: identity used twice

Evaluate $\int \cos^4 x \, dx$.

Step 1. Identify the powers: $\cos x$ has power 4 and $\sin x$ has power 0, both even, so use the half-angle identity $\cos^2 x = \frac{1 + \cos(2x)}{2}$.

Step 2. Write the fourth power as a square of a square: $\cos^4 x = (\cos^2 x)^2$.

Step 3. Apply the identity inside the parentheses:

$$\cos^4 x = \left(\frac{1 + \cos(2x)}{2} \right)^2.$$

Step 4. Square the fraction: numerator $(1 + \cos 2x)^2 = 1 + 2 \cos 2x + \cos^2 2x$ (using $(a + b)^2 = a^2 + 2ab + b^2$) and denominator $2^2 = 4$. So

$$\cos^4 x = \frac{1 + 2 \cos 2x + \cos^2 2x}{4}.$$

Step 5. The term $\cos^2 2x$ is again an even power, so apply the half-angle identity a second

time. Replace the angle x by $2x$ in $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$ to get $\cos^2 2x = \frac{1 + \cos 4x}{2}$.

Step 6. Substitute this into the numerator:

$$\cos^4 x = \frac{1}{4} \left(1 + 2 \cos 2x + \frac{1 + \cos 4x}{2} \right) = \frac{1}{4} + \frac{\cos 2x}{2} + \frac{1}{8} + \frac{\cos 4x}{8}.$$

Step 7. Combine the constants: $\frac{1}{4} + \frac{1}{8} = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$. So

$$\cos^4 x = \frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}.$$

Step 8. Integrate term by term. $\int \frac{3}{8} dx = \frac{3x}{8}$.

Step 9. $\int \frac{\cos 2x}{2} dx$: since $\int \cos(2x) dx = \frac{\sin 2x}{2}$, this equals $\frac{1}{2} \cdot \frac{\sin 2x}{2} = \frac{\sin 2x}{4}$.

Step 10. $\int \frac{\cos 4x}{8} dx$: since $\int \cos(4x) dx = \frac{\sin 4x}{4}$, this equals $\frac{1}{8} \cdot \frac{\sin 4x}{4} = \frac{\sin 4x}{32}$.

Step 11. Add the three results and the constant:

$$\frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C.$$

Step 12. Verify at one value. The derivative of the answer is $\frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$. At $x = 0$

this is $\frac{3}{8} + \frac{1}{2} + \frac{1}{8} = 1$, and $\cos^4 0 = 1$. At $x = \frac{\pi}{2}$ it is $\frac{3}{8} - \frac{1}{2} + \frac{1}{8} = 0$, and $\cos^4 \frac{\pi}{2} = 0$.

Answer. $\frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

EXAMPLE 4

Level: Intermediate. Secant and tangent with odd tangent power

Evaluate $\int \sec^4 x \tan^3 x dx$.

Step 1. Identify the powers: secant has power $n = 4$ (even) and tangent has power $m = 3$ (odd). We will use the derivatives $\frac{d}{dx}[\sec x] = \sec x \tan x$ and the identity $\tan^2 x + 1 = \sec^2 x$.

Step 2. Two rules apply here. Because the tangent power is odd, we can peel off one factor $\sec x \tan x$ (which is $d(\sec x)$) and use $u = \sec x$. (The even secant power would also allow $u = \tan x$; we follow the odd-tangent rule.)

Step 3. Split off the factor: $\sec^4 x \tan^3 x = \sec^3 x \tan^2 x \cdot (\sec x \tan x)$. Check: $\sec^3 \cdot \sec = \sec^4$ and $\tan^2 \cdot \tan = \tan^3$.

Step 4. Convert the remaining even tangent power with $\tan^2 x = \sec^2 x - 1$:

$$\sec^4 x \tan^3 x = \sec^3 x (\sec^2 x - 1) (\sec x \tan x).$$

Step 5. Substitute $u = \sec x$, so $du = \sec x \tan x dx$. Every factor is now in terms of u :

$$\int u^3(u^2 - 1) du.$$

Step 6. Expand: $u^3(u^2 - 1) = u^5 - u^3$.

Step 7. Apply the power rule: $\int (u^5 - u^3) du = \frac{u^6}{6} - \frac{u^4}{4} + C$.

Step 8. Back-substitute $u = \sec x$:

$$\frac{\sec^6 x}{6} - \frac{\sec^4 x}{4} + C.$$

Step 9. Verify by differentiating with the chain rule: $\frac{d}{dx} \left[\frac{\sec^6 x}{6} \right] = \sec^5 x \cdot \sec x \tan x =$

$$\sec^6 x \tan x \text{ and } \frac{d}{dx} \left[\frac{\sec^4 x}{4} \right] = \sec^3 x \cdot \sec x \tan x = \sec^4 x \tan x.$$

Step 10. Subtract: $\sec^6 x \tan x - \sec^4 x \tan x = \sec^4 x \tan x (\sec^2 x - 1) = \sec^4 x \tan x \cdot \tan^2 x = \sec^4 x \tan^3 x$, the original integrand.

Answer. $\frac{\sec^6 x}{6} - \frac{\sec^4 x}{4} + C$

EXAMPLE 5

Level: Advanced. Both powers even: sine squared times cosine squared

Evaluate $\int \sin^2 x \cos^2 x \, dx$.

Step 1. Identify the powers: $\sin x$ has power 2 and $\cos x$ has power 2, both even, so no factor can be peeled off for a substitution. Applying half-angle identities to both factors separately would give a product of two sums, which is messy, so first combine the factors.

Step 2. Recall the double-angle formula $\sin 2x = 2 \sin x \cos x$. Dividing by 2: $\sin x \cos x = \frac{\sin 2x}{2}$.

Step 3. Square both sides:

$$\sin^2 x \cos^2 x = (\sin x \cos x)^2 = \frac{\sin^2 2x}{4}.$$

Step 4. The new factor $\sin^2 2x$ is an even power of a single function, so use the half-angle identity $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ with $\theta = 2x$:

$$\sin^2 2x = \frac{1 - \cos 4x}{2}.$$

Step 5. Substitute:

$$\sin^2 x \cos^2 x = \frac{1}{4} \cdot \frac{1 - \cos 4x}{2} = \frac{1 - \cos 4x}{8}.$$

Step 6. Check the identity at $x = \frac{\pi}{4}$: the left side is $\left(\frac{\sqrt{2}}{2}\right)^2 \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{4}$ and the right side is $\frac{1 - \cos \pi}{8} = \frac{2}{8} = \frac{1}{4}$. They agree.

Step 7. Integrate:

$$\int \frac{1 - \cos 4x}{8} \, dx = \frac{1}{8} \int 1 \, dx - \frac{1}{8} \int \cos 4x \, dx.$$

Step 8. The first term is $\frac{x}{8}$. For the second, $\int \cos 4x \, dx = \frac{\sin 4x}{4}$ (substitution $w = 4x$, $dx = \frac{dw}{4}$), so the term is $\frac{1}{8} \cdot \frac{\sin 4x}{4} = \frac{\sin 4x}{32}$.

Step 9. Combine:

$$\frac{x}{8} - \frac{\sin 4x}{32} + C.$$

Step 10. Verify by differentiating: $\frac{d}{dx} \left[\frac{x}{8} - \frac{\sin 4x}{32} \right] = \frac{1}{8} - \frac{4 \cos 4x}{32} = \frac{1 - \cos 4x}{8} =$

$\sin^2 x \cos^2 x$, as required.

Answer. $\frac{x}{8} - \frac{\sin 4x}{32} + C$

EXAMPLE 6

Level: Intermediate. Even power of secant: the substitution $u = \tan x$

Evaluate $\int \sec^4 x \tan^2 x \, dx$.

Step 1. Identify the powers: secant has power 4 (even) and tangent has power 2 (even). The tangent power is not odd, so the odd-tangent rule ($u = \sec x$) does not apply. The secant power is even, so we use the even-secant rule with $u = \tan x$.

Step 2. The reason this works: $\frac{d}{dx}[\tan x] = \sec^2 x$, so a single factor $\sec^2 x \, dx$ becomes du .

Step 3. Peel off one factor $\sec^2 x$: $\sec^4 x = \sec^2 x \cdot \sec^2 x$.

Step 4. Convert the other $\sec^2 x$ into tangents with the identity $\sec^2 x = 1 + \tan^2 x$:

$$\sec^4 x \tan^2 x = (1 + \tan^2 x) \tan^2 x \cdot \sec^2 x.$$

Step 5. Substitute $u = \tan x$, so $du = \sec^2 x \, dx$. The integral becomes $\int (1 + u^2)u^2 \, du$.

Step 6. Expand: $(1 + u^2)u^2 = u^2 + u^4$.

Step 7. Apply the power rule: $\int (u^2 + u^4) \, du = \frac{u^3}{3} + \frac{u^5}{5} + C$.

Step 8. Back-substitute $u = \tan x$:

$$\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C.$$

Step 9. Verify by differentiating: $\frac{d}{dx} \left[\frac{\tan^3 x}{3} \right] = \tan^2 x \sec^2 x$ and $\frac{d}{dx} \left[\frac{\tan^5 x}{5} \right] = \tan^4 x \sec^2 x$.

Step 10. Add: $\tan^2 x \sec^2 x + \tan^4 x \sec^2 x = \tan^2 x \sec^2 x (1 + \tan^2 x) = \tan^2 x \sec^2 x \cdot \sec^2 x = \sec^4 x \tan^2 x$, the original integrand.

Step 11. Remark on where $\int \sec x \, dx$ enters: when the secant power is odd and the tangent power is even (for example $\int \sec x \, dx$ itself), neither rule applies. Multiplying by $\frac{\sec x + \tan x}{\sec x + \tan x}$ gives $\int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$, and with $u = \sec x + \tan x$, $du = (\sec x \tan x + \sec^2 x) \, dx$, the result is $\ln |\sec x + \tan x| + C$. It is not needed in this example because the secant power here is even.

Answer. $\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$

EXAMPLE 7*Level: Foundational. Long division, then integrate*

Evaluate

$$\int \frac{x^2 + 3x + 5}{x + 1} dx.$$

Step 1. Compare degrees. The numerator $x^2 + 3x + 5$ has degree 2 and the denominator $x + 1$ has degree 1. Partial fractions only work when the numerator has strictly smaller degree than the denominator, so we must first divide the numerator by the denominator (long division).

Step 2. Start the division. Divide the leading term of the numerator by the leading term of the denominator: $\frac{x^2}{x} = x$. This is the first term of the quotient.

Step 3. Multiply that term back by the whole divisor: $x(x + 1) = x^2 + x$. Subtract it from the numerator: $(x^2 + 3x + 5) - (x^2 + x) = 2x + 5$.

Step 4. Repeat with the new leading term: $\frac{2x}{x} = 2$. Multiply back: $2(x + 1) = 2x + 2$. Subtract: $(2x + 5) - (2x + 2) = 3$.

Step 5. The remainder 3 has degree 0, which is less than the degree 1 of the divisor, so the division stops. The quotient is $x + 2$ and the remainder is 3, that is $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$.

Step 6. Check the division by expanding: $(x + 1)(x + 2) = x^2 + 2x + x + 2 = x^2 + 3x + 2$, and adding the remainder gives $x^2 + 3x + 2 + 3 = x^2 + 3x + 5$, which is the original numerator.

Step 7. Divide both sides of $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$ by $x + 1$ to rewrite the integrand:

$$\frac{x^2 + 3x + 5}{x + 1} = x + 2 + \frac{3}{x + 1}.$$

Step 8. Split the integral into three pieces:

$$\int \left(x + 2 + \frac{3}{x + 1} \right) dx = \int x dx + \int 2 dx + 3 \int \frac{dx}{x + 1}.$$

Step 9. Integrate the first two pieces with the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$ and the constant rule: $\int x dx = \frac{x^2}{2}$ and $\int 2 dx = 2x$.

Step 10. For the third piece let $u = x + 1$, so $du = dx$. Then $\int \frac{dx}{x + 1} = \int \frac{du}{u} = \ln |u| = \ln |x + 1|$, using $\int \frac{du}{u} = \ln |u| + C$.

Step 11. Assemble the result:

$$\int \frac{x^2 + 3x + 5}{x + 1} dx = \frac{x^2}{2} + 2x + 3 \ln |x + 1| + C.$$

Step 12. Verify by differentiating: $\frac{d}{dx} \left[\frac{x^2}{2} + 2x + 3 \ln |x + 1| \right] = x + 2 + \frac{3}{x + 1} = \frac{(x + 2)(x + 1) + 3}{x + 1} = \frac{x^2 + 3x + 5}{x + 1}$, which is the original integrand.

Answer. $\frac{x^2}{2} + 2x + 3 \ln |x + 1| + C$

EXAMPLE 8*Level: Intermediate. Repeated linear factor*

Evaluate

$$\int \frac{2x + 6}{(x + 1)^2} dx.$$

Step 1. The numerator has degree 1 and the denominator $(x + 1)^2$ has degree 2, so no long division is needed and we can decompose directly.

Step 2. The denominator contains the repeated linear factor $(x + 1)^2$. A repeated factor needs one term for every power from 1 up to 2, so the shape is

$$\frac{2x + 6}{(x + 1)^2} = \frac{A}{x + 1} + \frac{B}{(x + 1)^2}.$$

Step 3. Multiply both sides by the common denominator $(x + 1)^2$ to clear the fractions:

$$2x + 6 = A(x + 1) + B.$$

Step 4. Find B by choosing the value of x that kills the A term, namely $x = -1$: $2(-1) + 6 = A(0) + B$, so $B = 4$.

Step 5. Find A by comparing the coefficients of x on both sides. The right side expands to $Ax + A + B$, whose coefficient of x is A ; the left side has coefficient 2. So $A = 2$.

Step 6. Check with the constant terms: $A + B = 2 + 4 = 6$, which matches the constant 6 on the left. The decomposition is

$$\frac{2x + 6}{(x + 1)^2} = \frac{2}{x + 1} + \frac{4}{(x + 1)^2}.$$

Step 7. Integrate term by term:

$$\int \frac{2x + 6}{(x + 1)^2} dx = 2 \int \frac{dx}{x + 1} + 4 \int \frac{dx}{(x + 1)^2}.$$

Step 8. Substitute $u = x + 1$, $du = dx$ in both integrals. The first is $2 \int \frac{du}{u} = 2 \ln |u| = 2 \ln |x + 1|$.

Step 9. The second is $4 \int u^{-2} du$. By the power rule $\int u^{-2} du = \frac{u^{-1}}{-1} = -\frac{1}{u}$, so it equals $-\frac{4}{u} = -\frac{4}{x + 1}$.

Step 10. Combine:

$$\int \frac{2x + 6}{(x + 1)^2} dx = 2 \ln |x + 1| - \frac{4}{x + 1} + C.$$

Step 11. Verify by differentiating: $\frac{d}{dx} \left[2 \ln |x + 1| - \frac{4}{x + 1} \right] = \frac{2}{x + 1} + \frac{4}{(x + 1)^2} = \frac{2(x + 1) + 4}{(x + 1)^2} = \frac{2x + 6}{(x + 1)^2}$, the original integrand.

Answer. $2 \ln |x + 1| - \frac{4}{x + 1} + C$

EXAMPLE 9*Level: Intermediate. Irreducible quadratic factor*

Evaluate

$$\int \frac{3x+1}{(x^2+1)(x-2)} dx.$$

Step 1. The numerator has degree 1 and the denominator has degree 3, so we decompose directly. The factor $x-2$ is a distinct linear factor. The factor x^2+1 cannot be factored over the real numbers (it has no real roots, since $x^2+1 \geq 1$), so it is an irreducible quadratic.

Step 2. An irreducible quadratic contributes a term with a linear numerator, and a linear factor contributes a constant numerator:

$$\frac{3x+1}{(x^2+1)(x-2)} = \frac{Ax+B}{x^2+1} + \frac{C}{x-2}.$$

Step 3. Multiply both sides by $(x^2+1)(x-2)$:

$$3x+1 = (Ax+B)(x-2) + C(x^2+1).$$

Step 4. Find C by substituting $x=2$, which makes $x-2=0$: $3(2)+1 = (2A+B)(0) + C(4+1)$, so $7 = 5C$ and $C = \frac{7}{5}$.

Step 5. Expand the right side fully: $(Ax+B)(x-2) = Ax^2 - 2Ax + Bx - 2B$, so

$$3x+1 = (A+C)x^2 + (B-2A)x + (C-2B).$$

Step 6. Match coefficients of x^2 : the left side has 0, so $A+C=0$ and $A=-C=-\frac{7}{5}$.

Step 7. Match the constant terms: $C-2B=1$, so $2B=C-1=\frac{7}{5}-1=\frac{2}{5}$ and $B=\frac{1}{5}$.

Step 8. Check with the coefficient of x : $B-2A=\frac{1}{5}+\frac{14}{5}=\frac{15}{5}=3$, which matches the 3 on the left. So

$$\frac{3x+1}{(x^2+1)(x-2)} = \frac{-\frac{7}{5}x+\frac{1}{5}}{x^2+1} + \frac{\frac{7}{5}}{x-2}.$$

Step 9. Split the first fraction into two and integrate each part:

$$\int \frac{3x+1}{(x^2+1)(x-2)} dx = -\frac{7}{5} \int \frac{x}{x^2+1} dx + \frac{1}{5} \int \frac{dx}{x^2+1} + \frac{7}{5} \int \frac{dx}{x-2}.$$

Step 10. For the first integral let $u = x^2+1$, so $du = 2x dx$ and $x dx = \frac{du}{2}$. Then

$$\int \frac{x}{x^2+1} dx = \frac{1}{2} \ln|u| = \frac{1}{2} \ln(x^2+1) \text{ (no absolute value needed since } x^2+1 > 0 \text{)}.$$

Step 11. The second integral is the standard form $\int \frac{du}{u^2+a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right)$ with $a=1$, giving $\arctan x$. The third is $\ln|x-2|$ by $u=x-2$.

Step 12. Assemble:

$$-\frac{7}{5} \cdot \frac{1}{2} \ln(x^2+1) + \frac{1}{5} \arctan x + \frac{7}{5} \ln|x-2| + C = -\frac{7}{10} \ln(x^2+1) + \frac{1}{5} \arctan x + \frac{7}{5} \ln|x-2| + C.$$

Step 13. Verify at $x=0$: the integrand is $\frac{1}{(1)(-2)} = -\frac{1}{2}$. The derivative of the answer at 0

$$\text{is } -\frac{7}{10} \cdot \frac{0}{1} + \frac{1}{5} \cdot \frac{1}{1} + \frac{7}{5} \cdot \frac{1}{-2} = \frac{1}{5} - \frac{7}{10} = -\frac{1}{2}, \text{ as required.}$$

Answer. $-\frac{7}{10} \ln(x^2+1) + \frac{1}{5} \arctan x + \frac{7}{5} \ln|x-2| + C$

EXAMPLE 10*Level: Advanced. Definite integral with an exact logarithm*

Evaluate

$$\int_0^1 \frac{4x+5}{(x+1)(x+2)} dx$$

and give the exact value as a single logarithm.

Step 1. The numerator has degree 1 and the denominator has degree 2, and the denominator is already factored into distinct linear factors, so the shape is

$$\frac{4x+5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}.$$

Step 2. Clear denominators by multiplying by $(x+1)(x+2)$:

$$4x+5 = A(x+2) + B(x+1).$$

Step 3. Substitute $x = -1$ (which kills the B term): $4(-1) + 5 = A(1) + B(0)$, so $A = 1$.**Step 4.** Substitute $x = -2$ (which kills the A term): $4(-2) + 5 = A(0) + B(-1)$, so $-3 = -B$ and $B = 3$.**Step 5.** Check by recombining: $1 \cdot (x+2) + 3(x+1) = x+2+3x+3 = 4x+5$, which matches. So

$$\frac{4x+5}{(x+1)(x+2)} = \frac{1}{x+1} + \frac{3}{x+2}.$$

Step 6. Find an antiderivative. Using $\int \frac{dx}{x+a} = \ln|x+a|$ (substitute $u = x+a$),

$$\int \left(\frac{1}{x+1} + \frac{3}{x+2} \right) dx = \ln|x+1| + 3\ln|x+2|.$$

Step 7. The integrand is continuous on $[0, 1]$ (the denominator vanishes only at $x = -1$ and $x = -2$), so the Fundamental Theorem of Calculus applies. On $[0, 1]$ both $x+1$ and $x+2$ are positive, so the absolute values can be dropped.**Step 8.** Evaluate at the upper limit $x = 1$: $\ln 2 + 3\ln 3$.**Step 9.** Evaluate at the lower limit $x = 0$: $\ln 1 + 3\ln 2 = 0 + 3\ln 2 = 3\ln 2$.**Step 10.** Subtract (upper minus lower): $\ln 2 + 3\ln 3 - 3\ln 2 = 3\ln 3 - 2\ln 2$.**Step 11.** Combine into one logarithm using $n \ln a = \ln a^n$ and $\ln a - \ln b = \ln \frac{a}{b}$: $3\ln 3 - 2\ln 2 = \ln 27 - \ln 4 = \ln \frac{27}{4}$.**Step 12.** Sanity check numerically: $3\ln 3 - 2\ln 2 \approx 3(1.0986) - 2(0.6931) = 3.2958 - 1.3863 = 1.9095$, and $\ln \frac{27}{4} = \ln 6.75 \approx 1.9095$. The integrand is between $\frac{5}{2}$ (at $x = 0$) and $\frac{9}{6} = 1.5$ (at $x = 1$) on an interval of length 1, so a value near 1.9 is reasonable.**Answer.** $\ln \frac{27}{4} = 3\ln 3 - 2\ln 2 \approx 1.9095$ **EXAMPLE 11***Level: Advanced. Weierstrass substitution with arctangent*

Evaluate

$$\int \frac{1}{2 + \cos x} dx.$$

Step 1. Choose the technique. The integrand is a rational function of $\cos x$ with no power

of sine or cosine to peel off and no secant or tangent structure, so the odd-power, even-power and sec/tan rules do not apply. The Weierstrass substitution $t = \tan\left(\frac{x}{2}\right)$ converts any rational function of $\sin x$ and $\cos x$ into a rational function of t .

Step 2. Derive $\cos x$ in terms of t . By the double-angle identity $\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)$, and by the Pythagorean identity $1 = \cos^2\left(\frac{x}{2}\right) + \sin^2\left(\frac{x}{2}\right)$. Write $\cos x = \frac{\cos^2(x/2) - \sin^2(x/2)}{\cos^2(x/2) + \sin^2(x/2)}$ and divide numerator and denominator by $\cos^2\left(\frac{x}{2}\right)$ to get

$$\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} = \frac{1 - t^2}{1 + t^2}.$$

Step 3. Find dx . From $t = \tan\left(\frac{x}{2}\right)$ we get $x = 2 \arctan t$, and since $\frac{d}{dt} \arctan t = \frac{1}{1 + t^2}$,

$$dx = \frac{2 dt}{1 + t^2}.$$

(This is valid for $-\pi < x < \pi$, where $\tan\left(\frac{x}{2}\right)$ is defined.)

Step 4. Substitute into the integral:

$$\int \frac{1}{2 + \frac{1 - t^2}{1 + t^2}} \cdot \frac{2 dt}{1 + t^2}.$$

Step 5. Simplify the denominator by writing 2 over the common denominator $1 + t^2$: $2 + \frac{1 - t^2}{1 + t^2} = \frac{2(1 + t^2) + (1 - t^2)}{1 + t^2} = \frac{2 + 2t^2 + 1 - t^2}{1 + t^2} = \frac{t^2 + 3}{1 + t^2}$.

Step 6. Dividing by a fraction means multiplying by its reciprocal, so the integrand becomes $\frac{1 + t^2}{t^2 + 3} \cdot \frac{2}{1 + t^2}$. The factor $1 + t^2$ cancels:

$$\int \frac{2 dt}{t^2 + 3}.$$

Step 7. Recognise the standard form $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$ with $u = t$ and $a = \sqrt{3}$ (since $3 = (\sqrt{3})^2$).

Step 8. Apply it:

$$\int \frac{2 dt}{t^2 + 3} = \frac{2}{\sqrt{3}} \arctan\left(\frac{t}{\sqrt{3}}\right) + C.$$

Step 9. Return to the original variable by replacing $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{1}{2 + \cos x} dx = \frac{2}{\sqrt{3}} \arctan\left(\frac{\tan(x/2)}{\sqrt{3}}\right) + C.$$

Step 10. Verify by differentiating with respect to x . Use $\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) = \frac{1 + t^2}{2}$ and the chain rule:

$$\frac{d}{dx} \left[\frac{2}{\sqrt{3}} \arctan\left(\frac{t}{\sqrt{3}}\right) \right] = \frac{2}{\sqrt{3}} \cdot \frac{1}{1 + t^2/3} \cdot \frac{1}{\sqrt{3}} \cdot \frac{1 + t^2}{2} = \frac{1 + t^2}{3 + t^2}.$$

Step 11. Compare with the integrand: $\frac{1}{2 + \cos x} = \frac{1 + t^2}{3 + t^2}$ from step 5 (the reciprocal of

$\frac{t^2 + 3}{1 + t^2}$). The two agree, so the antiderivative is correct. The substitution turned an integral with no obvious pattern into a routine arctangent integral.

Answer. $\frac{2}{\sqrt{3}} \arctan \left(\frac{\tan(x/2)}{\sqrt{3}} \right) + C$

EXAMPLE 12

Level: Challenge. Weierstrass substitution leading to a logarithm

Evaluate

$$\int \frac{1}{4 + 5 \sin x} dx.$$

Step 1. The integrand is a rational function of $\sin x$ alone with no power to peel off, so use the Weierstrass substitution $t = \tan \left(\frac{x}{2} \right)$, for which $\sin x = \frac{2t}{1 + t^2}$ and $dx = \frac{2 dt}{1 + t^2}$. (The identity for sine comes from $\sin x = 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)$ divided by $1 = \cos^2 \left(\frac{x}{2} \right) + \sin^2 \left(\frac{x}{2} \right)$, then dividing top and bottom by $\cos^2 \left(\frac{x}{2} \right)$.)

Step 2. Rewrite the denominator over the common denominator $1 + t^2$:

$$4 + 5 \sin x = 4 + \frac{10t}{1 + t^2} = \frac{4(1 + t^2) + 10t}{1 + t^2} = \frac{4t^2 + 10t + 4}{1 + t^2}.$$

Step 3. Substitute into the integral and take the reciprocal of the denominator:

$$\int \frac{1 + t^2}{4t^2 + 10t + 4} \cdot \frac{2 dt}{1 + t^2} = \int \frac{2 dt}{4t^2 + 10t + 4}.$$

Step 4. Cancel the factor $1 + t^2$ and divide numerator and denominator by 2:

$$\int \frac{dt}{2t^2 + 5t + 2}.$$

Step 5. Decide arctangent or logarithm by the discriminant of $2t^2 + 5t + 2$: $b^2 - 4ac = 25 - 16 = 9 > 0$. It is positive, so the quadratic factors over the real numbers and we expect a logarithm rather than an arctangent.

Step 6. Factor: $2t^2 + 5t + 2 = (2t + 1)(t + 2)$. Check by expanding: $(2t + 1)(t + 2) = 2t^2 + 4t + t + 2 = 2t^2 + 5t + 2$.

Step 7. Set up partial fractions:

$$\frac{1}{(2t + 1)(t + 2)} = \frac{A}{2t + 1} + \frac{B}{t + 2}, \quad \text{so} \quad 1 = A(t + 2) + B(2t + 1).$$

Step 8. Substitute $t = -2$: $1 = A(0) + B(-4 + 1) = -3B$, so $B = -\frac{1}{3}$.

Step 9. Substitute $t = -\frac{1}{2}$: $1 = A \left(\frac{3}{2} \right) + B(0)$, so $A = \frac{2}{3}$. Check by recombining: $\frac{2}{3}(t + 2) - \frac{1}{3}(2t + 1) = \frac{2t + 4 - 2t - 1}{3} = 1$.

Step 10. Integrate, using $\int \frac{dt}{at + b} = \frac{1}{a} \ln |at + b|$:

$$\int \left(\frac{2/3}{2t + 1} - \frac{1/3}{t + 2} \right) dt = \frac{2}{3} \cdot \frac{1}{2} \ln |2t + 1| - \frac{1}{3} \ln |t + 2| = \frac{1}{3} \ln |2t + 1| - \frac{1}{3} \ln |t + 2|.$$

Step 11. Combine with $\ln a - \ln b = \ln \frac{a}{b}$ and substitute back $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{dx}{4 + 5 \sin x} = \frac{1}{3} \ln \left| \frac{2 \tan(x/2) + 1}{\tan(x/2) + 2} \right| + C.$$

Step 12. Verify by differentiating. With $\frac{dt}{dx} = \frac{1+t^2}{2}$: $\frac{d}{dx} \left[\frac{1}{3} \ln \left| \frac{2t+1}{t+2} \right| \right] = \frac{1}{3} \left(\frac{2}{2t+1} - \frac{1}{t+2} \right) \frac{1+t^2}{2} = \frac{1}{(2t+1)(t+2)} \cdot \frac{1+t^2}{2} = \frac{1+t^2}{4t^2+10t+4}$, and this equals $\frac{1}{4+5 \sin x}$ by step 2.

Answer. $\frac{1}{3} \ln \left| \frac{2 \tan(x/2) + 1}{\tan(x/2) + 2} \right| + C$

Practice Problems

Worked-solutions version: every step of each solution is shown in green beneath its problem.

Trigonometric Integrals

1. Evaluate (a) $\int \sin^{10} x \cos x \, dx$ and (b) $\int \sin^{10} x \cos^7 x \, dx$. Identify which technique each needs.

SOLUTION

Step 1. Part (a). The integrand $\sin^{10} x \cos x$ contains $\cos x \, dx$, which is exactly the differential of $\sin x$. So the direct substitution $u = \sin x$ works with no rewriting.

Step 2. Set $u = \sin x$, so $du = \cos x \, dx$. Then $\sin^{10} x = u^{10}$ and the integral is $\int u^{10} \, du$.

Step 3. Apply the power rule: $\int u^{10} \, du = \frac{u^{11}}{11} + C$, so (a) equals $\frac{\sin^{11} x}{11} + C$.

Step 4. Part (b). The power of cosine is 7 (odd), so we peel off one factor: $\cos^7 x = \cos^6 x \cdot \cos x$. The single $\cos x$ will supply du .

Step 5. Write the remaining even power as a power of $\cos^2 x$ and use $\cos^2 x = 1 - \sin^2 x$ (from $\sin^2 x + \cos^2 x = 1$): $\cos^6 x = (\cos^2 x)^3 = (1 - \sin^2 x)^3$.

Step 6. The integral is $\int \sin^{10} x (1 - \sin^2 x)^3 \cos x \, dx$. Substitute $u = \sin x$, $du = \cos x \, dx$:

$$\int u^{10} (1 - u^2)^3 \, du.$$

Step 7. Expand $(1 - u^2)^3$ with $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$: $(1 - u^2)^3 = 1 - 3u^2 + 3u^4 - u^6$.

Step 8. Multiply by u^{10} : $u^{10} - 3u^{12} + 3u^{14} - u^{16}$.

Step 9. Integrate term by term: $\frac{u^{11}}{11} - \frac{3u^{13}}{13} + \frac{3u^{15}}{15} - \frac{u^{17}}{17} + C$, and $\frac{3}{15} = \frac{1}{5}$.

Step 10. Back-substitute $u = \sin x$: (b) equals $\frac{\sin^{11} x}{11} - \frac{3 \sin^{13} x}{13} + \frac{\sin^{15} x}{5} - \frac{\sin^{17} x}{17} + C$.

Step 11. Verify (a): $\frac{d}{dx} \left[\frac{\sin^{11} x}{11} \right] = \sin^{10} x \cos x$. Verify (b) by differentiating each term: each gives $\sin^{k-1} x \cos x$ times its coefficient times k over k , and the sum is $\cos x (\sin^{10} x - 3 \sin^{12} x + 3 \sin^{14} x - \sin^{16} x) = \cos x \sin^{10} x (1 - \sin^2 x)^3$.

Answer. (a) $\frac{\sin^{11} x}{11} + C$ (substitution $u = \sin x$). (b) $\frac{\sin^{11} x}{11} - \frac{3 \sin^{13} x}{13} + \frac{\sin^{15} x}{5} - \frac{\sin^{17} x}{17} + C$ (peel off one cosine, odd power).

2. Evaluate $\int \sec^2 x \tan^3 x \, dx$.

SOLUTION

Step 1. Identify the pattern: the integrand contains $\sec^2 x$, and $\frac{d}{dx} [\tan x] = \sec^2 x$.

So $\sec^2 x \, dx$ is the differential of $\tan x$, and the rest of the integrand, $\tan^3 x$, is a function of $\tan x$.

Step 2. Choose the substitution $u = \tan x$.

Step 3. Differentiate: $du = \sec^2 x \, dx$.

Step 4. Replace $\tan^3 x$ by u^3 and $\sec^2 x \, dx$ by du : the integral becomes $\int u^3 \, du$.

Step 5. Apply the power rule $\int u^k \, du = \frac{u^{k+1}}{k+1}$ with $k = 3$: $\frac{u^4}{4} + C$.

Step 6. Back-substitute $u = \tan x$: $\frac{\tan^4 x}{4} + C$.

Step 7. Verify by the chain rule: $\frac{d}{dx} \left[\frac{\tan^4 x}{4} \right] = \frac{4 \tan^3 x \cdot \sec^2 x}{4} = \sec^2 x \tan^3 x$, the original integrand.

Answer. $\frac{\tan^4 x}{4} + C$

3. Evaluate $\int \tan^3 x \, dx$.

SOLUTION

Step 1. The integrand has no secant factor, so the sec/tan rules do not apply directly. Instead create a $\sec^2 x$ factor using the identity $\tan^2 x = \sec^2 x - 1$ (from $\tan^2 x + 1 = \sec^2 x$).

Step 2. Split off one tangent: $\tan^3 x = \tan x \cdot \tan^2 x$.

Step 3. Replace $\tan^2 x$: $\tan^3 x = \tan x(\sec^2 x - 1)$.

Step 4. Distribute: $\tan x(\sec^2 x - 1) = \tan x \sec^2 x - \tan x$. So

$$\int \tan^3 x \, dx = \int \tan x \sec^2 x \, dx - \int \tan x \, dx.$$

Step 5. First integral: let $u = \tan x$, $du = \sec^2 x \, dx$. Then $\int \tan x \sec^2 x \, dx = \int u \, du = \frac{u^2}{2} = \frac{\tan^2 x}{2}$.

Step 6. Second integral: write $\tan x = \frac{\sin x}{\cos x}$ and let $w = \cos x$, $dw = -\sin x \, dx$.

$$\text{Then } \int \frac{\sin x}{\cos x} \, dx = - \int \frac{dw}{w} = -\ln |w| = -\ln |\cos x|.$$

Step 7. Rewrite using $-\ln |\cos x| = \ln \left| \frac{1}{\cos x} \right| = \ln |\sec x|$. So $\int \tan x \, dx = \ln |\sec x| + C$.

Step 8. Combine, remembering the minus sign in front of the second integral:

$$\frac{\tan^2 x}{2} - \ln |\sec x| + C.$$

Step 9. Verify: $\frac{d}{dx} \left[\frac{\tan^2 x}{2} \right] = \tan x \sec^2 x$ and $\frac{d}{dx} [\ln |\sec x|] = \frac{\sec x \tan x}{\sec x} = \tan x$.

Step 10. The difference is $\tan x \sec^2 x - \tan x = \tan x(\sec^2 x - 1) = \tan x \cdot \tan^2 x = \tan^3 x$, as required.

Answer. $\frac{\tan^2 x}{2} - \ln |\sec x| + C$

4. Evaluate $\int_0^{\pi/2} \cos^3 x \, dx$.

SOLUTION

Step 1. The power of cosine is 3 (odd), so peel off one factor: $\cos^3 x = \cos^2 x \cdot \cos x$.

Step 2. Convert the even power with $\cos^2 x = 1 - \sin^2 x$: $\cos^3 x = (1 - \sin^2 x) \cos x$.

Step 3. Choose $u = \sin x$, so $du = \cos x \, dx$.

Step 4. Change the limits of integration to match the new variable. When $x = 0$: $u = \sin 0 = 0$. When $x = \frac{\pi}{2}$: $u = \sin \frac{\pi}{2} = 1$.

Step 5. Rewrite the definite integral entirely in u :

$$\int_0^1 (1 - u^2) \, du.$$

Step 6. Find an antiderivative: $\int (1 - u^2) du = u - \frac{u^3}{3}$.

Step 7. Evaluate at the upper limit $u = 1$: $1 - \frac{1}{3} = \frac{2}{3}$.

Step 8. Evaluate at the lower limit $u = 0$: $0 - 0 = 0$.

Step 9. Subtract (upper minus lower): $\frac{2}{3} - 0 = \frac{2}{3}$.

Step 10. Sanity check: $\cos^3 x$ is between 0 and 1 on $\left[0, \frac{\pi}{2}\right]$, an interval of length about 1.57, and $\cos^3 x \leq \cos x$ whose integral is 1. The value $\frac{2}{3} \approx 0.667$ is below 1 and positive, which is consistent.

Answer. $\frac{2}{3}$

5. Evaluate $\int_0^\pi \cos^2 x \, dx$.

SOLUTION

Step 1. Both powers are even ($\cos^2 x$ and $\sin^0 x$), so use the half-angle identity $\cos^2 x = \frac{1 + \cos 2x}{2}$ (from $\cos 2x = 2\cos^2 x - 1$).

Step 2. Rewrite: $\int_0^\pi \cos^2 x \, dx = \int_0^\pi \frac{1 + \cos 2x}{2} \, dx$.

Step 3. Find an antiderivative of $\frac{1 + \cos 2x}{2}$: the term $\frac{1}{2}$ gives $\frac{x}{2}$, and $\frac{\cos 2x}{2}$ gives $\frac{1}{2} \cdot \frac{\sin 2x}{2} = \frac{\sin 2x}{4}$ (substitution $w = 2x$, $dx = \frac{dw}{2}$).

Step 4. So $F(x) = \frac{x}{2} + \frac{\sin 2x}{4}$. No substitution was made in the definite integral itself, so the limits stay 0 and π .

Step 5. Evaluate at the upper limit: $F(\pi) = \frac{\pi}{2} + \frac{\sin 2\pi}{4} = \frac{\pi}{2} + 0 = \frac{\pi}{2}$.

Step 6. Evaluate at the lower limit: $F(0) = 0 + \frac{\sin 0}{4} = 0$.

Step 7. Subtract: $F(\pi) - F(0) = \frac{\pi}{2}$.

Step 8. Cross-check by symmetry: $\sin^2 x + \cos^2 x = 1$, so $\int_0^\pi \sin^2 x \, dx + \int_0^\pi \cos^2 x \, dx = \pi$. Over a full half-period the two integrals are equal, so each is $\frac{\pi}{2}$, in agreement.

Answer. $\frac{\pi}{2}$

6. Evaluate $\int \tan^4 x \, dx$.

SOLUTION

Step 1. There is no secant factor, so create one with $\tan^2 x = \sec^2 x - 1$. Split $\tan^4 x = \tan^2 x \cdot \tan^2 x$.

Step 2. Replace only one of the two factors: $\tan^4 x = \tan^2 x(\sec^2 x - 1)$.

Step 3. Distribute: $\tan^4 x = \tan^2 x \sec^2 x - \tan^2 x$.

Step 4. Apply the identity once more to the leftover $\tan^2 x$: $\tan^2 x = \sec^2 x - 1$, so $\tan^4 x = \tan^2 x \sec^2 x - \sec^2 x + 1$.

Step 5. Integrate the first term with $u = \tan x$, $du = \sec^2 x \, dx$:

$$\int \tan^2 x \sec^2 x \, dx = \int u^2 \, du = \frac{u^3}{3} = \frac{\tan^3 x}{3}.$$

Step 6. Integrate the second term using $\frac{d}{dx}[\tan x] = \sec^2 x$: $\int \sec^2 x \, dx = \tan x$, so with its minus sign this contributes $-\tan x$.

Step 7. Integrate the third term: $\int 1 \, dx = x$.

Step 8. Add the pieces:

$$\frac{\tan^3 x}{3} - \tan x + x + C.$$

Step 9. Verify: the derivative is $\tan^2 x \sec^2 x - \sec^2 x + 1 = \sec^2 x(\tan^2 x - 1) + 1$. Using $\sec^2 x = \tan^2 x + 1$ this is $(\tan^2 x + 1)(\tan^2 x - 1) + 1 = \tan^4 x - 1 + 1 = \tan^4 x$, as required.

Answer. $\frac{\tan^3 x}{3} - \tan x + x + C$

7. Evaluate $\int \sin^4 x \, dx$. (Both powers are even, so apply the half-angle identity twice.)

SOLUTION

Step 1. Both exponents are even ($\sin^4 x$, $\cos^0 x$), so use the half-angle identity $\sin^2 x = \frac{1 - \cos 2x}{2}$ (from $\cos 2x = 1 - 2\sin^2 x$).

Step 2. Write $\sin^4 x = (\sin^2 x)^2 = \left(\frac{1 - \cos 2x}{2}\right)^2$.

Step 3. Square, using $(a - b)^2 = a^2 - 2ab + b^2$:

$$\sin^4 x = \frac{1 - 2\cos 2x + \cos^2 2x}{4}.$$

Step 4. The term $\cos^2 2x$ is again an even power. Use $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$ with $\theta = 2x$: $\cos^2 2x = \frac{1 + \cos 4x}{2}$.

Step 5. Substitute and distribute the $\frac{1}{4}$:

$$\sin^4 x = \frac{1}{4} - \frac{\cos 2x}{2} + \frac{1}{8} + \frac{\cos 4x}{8}.$$

Step 6. Combine constants: $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$. So $\sin^4 x = \frac{3}{8} - \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$.

Step 7. Integrate term by term: $\int \frac{3}{8} dx = \frac{3x}{8}$; $\int -\frac{\cos 2x}{2} dx = -\frac{1}{2} \cdot \frac{\sin 2x}{2} = -\frac{\sin 2x}{4}$; $\int \frac{\cos 4x}{8} dx = \frac{1}{8} \cdot \frac{\sin 4x}{4} = \frac{\sin 4x}{32}$.

Step 8. Combine:

$$\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C.$$

Step 9. Verify: the derivative is $\frac{3}{8} - \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$. At $x = \frac{\pi}{2}$ this is $\frac{3}{8} + \frac{1}{2} + \frac{1}{8} = 1 = \sin^4 \frac{\pi}{2}$, and at $x = 0$ it is $\frac{3}{8} - \frac{1}{2} + \frac{1}{8} = 0 = \sin^4 0$.

Answer. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

8. State which substitution ($u = \tan x$ or $u = \sec x$) is appropriate for $\int \sec^5 x \tan^3 x dx$ and explain why, using the rules for $\int \sec^n x \tan^m x dx$. Then evaluate the integral.

SOLUTION

Step 1. Test the even-secant rule: it needs n (the secant power) to be even so that a factor $\sec^2 x$ can be peeled off for $u = \tan x$. Here $n = 5$ is odd, so that rule fails.

Step 2. Test the odd-tangent rule: it needs m (the tangent power) to be odd so that one factor $\sec x \tan x$ can be peeled off for $u = \sec x$. Here $m = 3$ is odd and $n \geq 1$, so this rule applies. The substitution is $u = \sec x$.

Step 3. Peel off $\sec x \tan x$: $\sec^5 x \tan^3 x = \sec^4 x \tan^2 x \cdot (\sec x \tan x)$. Check: $\sec^4 \cdot \sec = \sec^5$ and $\tan^2 \cdot \tan = \tan^3$.

Step 4. Convert the even tangent power using $\tan^2 x = \sec^2 x - 1$: $\sec^5 x \tan^3 x = \sec^4 x (\sec^2 x - 1)(\sec x \tan x)$.

Step 5. Substitute $u = \sec x$, $du = \sec x \tan x dx$:

$$\int u^4(u^2 - 1) du.$$

Step 6. Expand: $u^4(u^2 - 1) = u^6 - u^4$.

Step 7. Integrate with the power rule: $\frac{u^7}{7} - \frac{u^5}{5} + C$.

Step 8. Back-substitute $u = \sec x$:

$$\frac{\sec^7 x}{7} - \frac{\sec^5 x}{5} + C.$$

Step 9. Verify: $\frac{d}{dx} \left[\frac{\sec^7 x}{7} \right] = \sec^6 x \cdot \sec x \tan x = \sec^7 x \tan x$ and

$$\frac{d}{dx} \left[\frac{\sec^5 x}{5} \right] = \sec^5 x \tan x.$$

Step 10. Subtract: $\sec^7 x \tan x - \sec^5 x \tan x = \sec^5 x \tan x (\sec^2 x - 1) = \sec^5 x \tan^3 x$, the original integrand.

Answer. Use $u = \sec x$ (tangent power 3 is odd; secant power 5 is odd so $u = \tan x$ fails). Result: $\frac{\sec^7 x}{7} - \frac{\sec^5 x}{5} + C$

9. Evaluate $\int \sin^2 x \cos^4 x dx$.

SOLUTION

Step 1. Both exponents are even, so no factor can be peeled off for a substitution. Combine part of the product first: $\sin^2 x \cos^4 x = (\sin x \cos x)^2 \cos^2 x$.

Step 2. Use $\sin 2x = 2 \sin x \cos x$, so $\sin x \cos x = \frac{\sin 2x}{2}$ and $(\sin x \cos x)^2 = \frac{\sin^2 2x}{4}$.

Step 3. Use $\cos^2 x = \frac{1 + \cos 2x}{2}$ for the remaining factor, so

$$\sin^2 x \cos^4 x = \frac{\sin^2 2x}{4} \cdot \frac{1 + \cos 2x}{2} = \frac{\sin^2 2x (1 + \cos 2x)}{8}.$$

Step 4. Distribute: $\sin^2 x \cos^4 x = \frac{\sin^2 2x}{8} + \frac{\sin^2 2x \cos 2x}{8}$.

Step 5. First term: use $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ with $\theta = 2x$, giving $\sin^2 2x = \frac{1 - \cos 4x}{2}$. So $\frac{\sin^2 2x}{8} = \frac{1 - \cos 4x}{16}$.

Step 6. Integrate it: $\int \frac{1 - \cos 4x}{16} dx = \frac{x}{16} - \frac{1}{16} \cdot \frac{\sin 4x}{4} = \frac{x}{16} - \frac{\sin 4x}{64}$.

Step 7. Second term: $\int \frac{\sin^2 2x \cos 2x}{8} dx$. It contains $\cos 2x$ (the derivative of $\sin 2x$ up to a constant), so let $w = \sin 2x$, $dw = 2 \cos 2x dx$, so $\cos 2x dx = \frac{dw}{2}$.

Step 8. Substitute: $\frac{1}{8} \int w^2 \cdot \frac{dw}{2} = \frac{1}{16} \cdot \frac{w^3}{3} = \frac{w^3}{48} = \frac{\sin^3 2x}{48}$.

Step 9. Add both results and the constant:

$$\frac{x}{16} - \frac{\sin 4x}{64} + \frac{\sin^3 2x}{48} + C.$$

Step 10. Verify at $x = \frac{\pi}{4}$: the integrand is $\sin^2 \frac{\pi}{4} \cos^4 \frac{\pi}{4} = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. The derivative of the answer is $\frac{1}{16} - \frac{\cos 4x}{16} + \frac{\sin^2 2x \cos 2x}{8}$, which at $x = \frac{\pi}{4}$ is $\frac{1}{16} + \frac{1}{16} + 0 = \frac{1}{8}$. They agree.

Answer. $\frac{x}{16} - \frac{\sin 4x}{64} + \frac{\sin^3 2x}{48} + C$

10. A student claims that $\int \sin^3 x \, dx = -\cos x - \frac{\cos^3 x}{3} + C$, using the step $\sin^3 x = \sin x (1 + \cos^2 x)$. Find the error and give the correct result.

SOLUTION

Step 1. Examine the student's step $\sin^3 x = \sin x (1 + \cos^2 x)$. It would need $\sin^2 x = 1 + \cos^2 x$.

Step 2. The correct Pythagorean identity is $\sin^2 x + \cos^2 x = 1$, so $\sin^2 x = 1 - \cos^2 x$. The student used a plus sign where a minus sign belongs.

Step 3. Test the student's claim numerically at $x = \frac{\pi}{4}$: $\sin^2 \frac{\pi}{4} = \frac{1}{2}$ but $1 + \cos^2 \frac{\pi}{4} = 1 + \frac{1}{2} = \frac{3}{2}$. These are not equal, so the step is false.

Step 4. Redo the problem. The sine power 3 is odd, so peel off one factor: $\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x) \sin x$.

Step 5. Distribute: $\sin^3 x = \sin x - \cos^2 x \sin x$, so $\int \sin^3 x \, dx = \int \sin x \, dx - \int \cos^2 x \sin x \, dx$.

Step 6. First integral: $\int \sin x \, dx = -\cos x$.

Step 7. Second integral: let $u = \cos x$, $du = -\sin x \, dx$. Then $\int \cos^2 x \sin x \, dx = -\int u^2 \, du = -\frac{u^3}{3} = -\frac{\cos^3 x}{3}$.

Step 8. Combine, remembering the subtraction: $-\cos x - \left(-\frac{\cos^3 x}{3}\right) = -\cos x + \frac{\cos^3 x}{3}$.

Step 9. Verify the correct answer: $\frac{d}{dx} \left[-\cos x + \frac{\cos^3 x}{3} \right] = \sin x + \frac{3 \cos^2 x (-\sin x)}{3} = \sin x - \cos^2 x \sin x = \sin x (1 - \cos^2 x) = \sin^3 x$.

Step 10. Verify that the student's answer fails: its derivative is $\sin x + \cos^2 x \sin x = \sin x(1 + \cos^2 x)$. At $x = \frac{\pi}{4}$ this is $\frac{\sqrt{2}}{2} \cdot \frac{3}{2} \approx 1.061$, while $\sin^3 \frac{\pi}{4} \approx 0.354$.

Answer. The error is $\sin^2 x = 1 + \cos^2 x$ (should be $1 - \cos^2 x$). Correct result:

$$-\cos x + \frac{\cos^3 x}{3} + C$$

Partial Fractions

11. Factor the denominator completely and evaluate

$$\int \frac{1}{x^2 + 3x} dx.$$

SOLUTION

Step 1. The numerator has degree 0 and the denominator degree 2, so no long division is needed. Factor the denominator by taking out the common factor x : $x^2 + 3x = x(x + 3)$.

Step 2. Both factors are distinct linear factors, so each gets one constant-numerator term:

$$\frac{1}{x(x + 3)} = \frac{A}{x} + \frac{B}{x + 3}.$$

Step 3. Multiply both sides by $x(x + 3)$:

$$1 = A(x + 3) + Bx.$$

Step 4. Substitute $x = 0$ (kills the B term): $1 = A(3) + 0$, so $A = \frac{1}{3}$.

Step 5. Substitute $x = -3$ (kills the A term): $1 = A(0) + B(-3)$, so $B = -\frac{1}{3}$.

Step 6. Check by recombining: $\frac{1}{3}(x + 3) - \frac{1}{3}x = \frac{x + 3 - x}{3} = 1$, which matches the left side. So

$$\frac{1}{x^2 + 3x} = \frac{1}{3} \cdot \frac{1}{x} - \frac{1}{3} \cdot \frac{1}{x + 3}.$$

Step 7. Integrate term by term using $\int \frac{du}{u} = \ln |u|$ (with $u = x$ and $u = x + 3$, $du = dx$):

$$\int \frac{1}{x^2 + 3x} dx = \frac{1}{3} \ln |x| - \frac{1}{3} \ln |x + 3| + C.$$

Step 8. Optionally combine with $\ln a - \ln b = \ln \frac{a}{b}$: the result is $\frac{1}{3} \ln \left| \frac{x}{x + 3} \right| + C$.

Step 9. Verify by differentiating: $\frac{1}{3} \cdot \frac{1}{x} - \frac{1}{3} \cdot \frac{1}{x + 3} = \frac{1}{3} \cdot \frac{(x + 3) - x}{x(x + 3)} = \frac{1}{3} \cdot \frac{3}{x(x + 3)} = \frac{1}{x^2 + 3x}$.

Answer. $\frac{1}{3} \ln |x| - \frac{1}{3} \ln |x + 3| + C = \frac{1}{3} \ln \left| \frac{x}{x + 3} \right| + C$

12. Factor the denominator completely and evaluate

$$\int \frac{1}{x^3 - x} dx.$$

SOLUTION

Step 1. The numerator has degree 0 and the denominator degree 3, so no long division is needed. Take out the common factor x : $x^3 - x = x(x^2 - 1)$.

Step 2. Do not stop there. The factor $x^2 - 1$ is a difference of squares, $a^2 - b^2 = (a - b)(a + b)$, so $x^2 - 1 = (x - 1)(x + 1)$ and

$$x^3 - x = x(x - 1)(x + 1).$$

Step 3. There are three distinct linear factors, so three terms are needed:

$$\frac{1}{x(x - 1)(x + 1)} = \frac{A}{x} + \frac{B}{x - 1} + \frac{C}{x + 1}.$$

Step 4. Multiply both sides by $x(x - 1)(x + 1)$:

$$1 = A(x - 1)(x + 1) + Bx(x + 1) + Cx(x - 1).$$

Step 5. Substitute $x = 0$: $1 = A(-1)(1) + 0 + 0$, so $A = -1$.

Step 6. Substitute $x = 1$: $1 = 0 + B(1)(2) + 0$, so $B = \frac{1}{2}$.

Step 7. Substitute $x = -1$: $1 = 0 + 0 + C(-1)(-2) = 2C$, so $C = \frac{1}{2}$.

Step 8. Check by comparing the x^2 coefficients: expanding gives $(A + B + C)x^2 + (B - C)x - A$, and $A + B + C = -1 + \frac{1}{2} + \frac{1}{2} = 0$ (left side has no x^2), $B - C = 0$ (no x term), $-A = 1$ (the constant). All three match.

Step 9. Integrate each term with $\int \frac{du}{u} = \ln |u|$:

$$\int \left(-\frac{1}{x} + \frac{1/2}{x - 1} + \frac{1/2}{x + 1} \right) dx = -\ln |x| + \frac{1}{2} \ln |x - 1| + \frac{1}{2} \ln |x + 1| + C.$$

Step 10. Verify at $x = 2$: the integrand is $\frac{1}{8 - 2} = \frac{1}{6}$, and the derivative of the answer is $-\frac{1}{2} + \frac{1}{2} \cdot \frac{1}{1} + \frac{1}{2} \cdot \frac{1}{3} = -\frac{1}{2} + \frac{1}{2} + \frac{1}{6} = \frac{1}{6}$.

Answer. $-\ln |x| + \frac{1}{2} \ln |x - 1| + \frac{1}{2} \ln |x + 1| + C$

13. Evaluate

$$\int \frac{5x + 1}{2x^2 + 5x - 3} dx.$$

SOLUTION

Step 1. The numerator has degree 1 and the denominator degree 2, so no long division is needed. First factor the denominator, whose leading coefficient is 2 rather than 1.

Step 2. Use the product-of-roots method: multiply the leading coefficient and constant, $2 \cdot (-3) = -6$, and find two numbers with product -6 and sum 5 (the middle coefficient). They are 6 and -1 .

Step 3. Split the middle term: $2x^2 + 5x - 3 = 2x^2 + 6x - x - 3$. Group: $2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3)$. Check by expanding: $(2x - 1)(x + 3) = 2x^2 + 6x - x - 3 = 2x^2 + 5x - 3$.

Step 4. Two distinct linear factors, so

$$\frac{5x + 1}{(2x - 1)(x + 3)} = \frac{A}{2x - 1} + \frac{B}{x + 3}.$$

Step 5. Multiply by $(2x - 1)(x + 3)$:

$$5x + 1 = A(x + 3) + B(2x - 1).$$

Step 6. Substitute $x = \frac{1}{2}$ (makes $2x - 1 = 0$): $5 \cdot \frac{1}{2} + 1 = \frac{7}{2} = A \left(\frac{7}{2} \right)$, so $A = 1$.

Step 7. Substitute $x = -3$ (makes $x + 3 = 0$): $5(-3) + 1 = -14 = B(-6 - 1) = -7B$, so $B = 2$.

Step 8. Check by recombining: $1 \cdot (x + 3) + 2(2x - 1) = x + 3 + 4x - 2 = 5x + 1$. So

$$\frac{5x + 1}{2x^2 + 5x - 3} = \frac{1}{2x - 1} + \frac{2}{x + 3}.$$

Step 9. Integrate the first term with $u = 2x - 1$, $du = 2 dx$, so $dx = \frac{du}{2}$:

$\int \frac{dx}{2x - 1} = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln |2x - 1|$. The factor $\frac{1}{2}$ comes from the leading coefficient 2 of the linear factor.

Step 10. Integrate the second term with $u = x + 3$: $2 \int \frac{dx}{x + 3} = 2 \ln |x + 3|$.

Step 11. Combine:

$$\int \frac{5x + 1}{2x^2 + 5x - 3} dx = \frac{1}{2} \ln |2x - 1| + 2 \ln |x + 3| + C.$$

Step 12. Verify by differentiating: $\frac{d}{dx} \left[\frac{1}{2} \ln |2x - 1| \right] = \frac{1}{2} \cdot \frac{2}{2x - 1} = \frac{1}{2x - 1}$ and

$\frac{d}{dx} [2 \ln |x + 3|] = \frac{2}{x + 3}$, whose sum is the decomposition above.

Answer. $\frac{1}{2} \ln |2x - 1| + 2 \ln |x + 3| + C$

14. A student evaluates $\int \frac{3x + 1}{(x - 1)(x + 2)} dx$ as follows. They write $\frac{3x + 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$, then use the cover-up shortcut: at $x = 1$ they compute $A = 3(1) + 1 = 4$,

and at $x = -2$ they compute $B = 3(-2) + 1 = -5$. They conclude that the integral equals $4 \ln |x - 1| - 5 \ln |x + 2| + C$. Find the error and give the correct result.

SOLUTION

Step 1. Test the claim first. Pick $x = 0$, where the integrand is $\frac{3(0) + 1}{(0 - 1)(0 + 2)} = \frac{1}{-2} = -\frac{1}{2}$.

Step 2. The derivative of the student answer at $x = 0$ is $\frac{4}{0 - 1} - \frac{5}{0 + 2} = -4 - \frac{5}{2} = -\frac{13}{2}$. This is not $-\frac{1}{2}$, so the claimed antiderivative is wrong.

Step 3. Locate the error. The decomposition shape is correct. The error is in the value of the coefficients. Clearing denominators gives

$$3x + 1 = A(x + 2) + B(x - 1).$$

Step 4. Setting $x = 1$ in this equation gives $3(1) + 1 = A(1 + 2) + B(0)$, that is $4 = 3A$. The student wrote $A = 4$ and forgot to divide by the value $x + 2 = 3$ of the remaining factor.

Step 5. Setting $x = -2$ gives $3(-2) + 1 = A(0) + B(-2 - 1)$, that is $-5 = -3B$. The student wrote $B = -5$, forgetting both to divide by the value $x - 1 = -3$ of the remaining factor and the resulting sign.

Step 6. Correct values: $A = \frac{4}{3}$ and $B = \frac{-5}{-3} = \frac{5}{3}$.

Step 7. Check by recombining: $\frac{4}{3}(x+2) + \frac{5}{3}(x-1) = \frac{4x+8+5x-5}{3} = \frac{9x+3}{3} = 3x+1$, which matches the numerator. (The student version would give $4(x+2) - 5(x-1) = -x+13 \neq 3x+1$.)

Step 8. So

$$\frac{3x+1}{(x-1)(x+2)} = \frac{4/3}{x-1} + \frac{5/3}{x+2}.$$

Step 9. Integrate each term with $\int \frac{du}{u} = \ln |u|$ ($u = x - 1$ and $u = x + 2$, $du = dx$):

$$\int \frac{3x+1}{(x-1)(x+2)} dx = \frac{4}{3} \ln |x-1| + \frac{5}{3} \ln |x+2| + C.$$

Step 10. Verify at $x = 0$: the derivative of the correct answer is $\frac{4}{3} \cdot \frac{1}{-1} + \frac{5}{3} \cdot \frac{1}{2} = -\frac{4}{3} + \frac{5}{6} = -\frac{8}{6} + \frac{5}{6} = -\frac{1}{2}$, which equals the integrand value found in the first step.

Answer. Error: the cover-up values were not divided by the remaining factor ($A = \frac{4}{3}$, $B = \frac{5}{3}$, not 4 and -5). Correct result: $\frac{4}{3} \ln |x - 1| + \frac{5}{3} \ln |x + 2| + C$

15. Evaluate the definite integral

$$\int_2^3 \frac{1}{x^2 - 1} dx$$

and give the exact value as a single logarithm.

SOLUTION

Step 1. The numerator has degree 0 and the denominator degree 2, so no long division is needed. Factor with the difference of squares $a^2 - b^2 = (a - b)(a + b)$: $x^2 - 1 = (x - 1)(x + 1)$.

Step 2. Set up two terms:

$$\frac{1}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1}, \quad 1 = A(x + 1) + B(x - 1).$$

Step 3. Substitute $x = 1$: $1 = A(2) + 0$, so $A = \frac{1}{2}$. Substitute $x = -1$: $1 = 0 + B(-2)$, so $B = -\frac{1}{2}$.

Step 4. Check by recombining: $\frac{1}{2}(x + 1) - \frac{1}{2}(x - 1) = \frac{x + 1 - x + 1}{2} = 1$. So

$$\frac{1}{x^2 - 1} = \frac{1}{2} \cdot \frac{1}{x - 1} - \frac{1}{2} \cdot \frac{1}{x + 1}.$$

Step 5. Find an antiderivative using $\int \frac{du}{u} = \ln |u|$: $F(x) = \frac{1}{2} \ln |x - 1| - \frac{1}{2} \ln |x + 1| = \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right|$.

Step 6. The integrand is continuous on $[2, 3]$ (the denominator vanishes only at $x = \pm 1$), and on this interval $x - 1 > 0$ and $x + 1 > 0$, so the absolute values may be dropped.

Step 7. Upper limit $x = 3$: $F(3) = \frac{1}{2} \ln \frac{2}{4} = \frac{1}{2} \ln \frac{1}{2}$.

Step 8. Lower limit $x = 2$: $F(2) = \frac{1}{2} \ln \frac{1}{3}$.

Step 9. Subtract: $F(3) - F(2) = \frac{1}{2} \left(\ln \frac{1}{2} - \ln \frac{1}{3} \right) = \frac{1}{2} \ln \left(\frac{1}{2} \div \frac{1}{3} \right) = \frac{1}{2} \ln \frac{3}{2}$, using $\ln a - \ln b = \ln \frac{a}{b}$.

Step 10. Numerical sanity check: $\frac{1}{2} \ln 1.5 \approx \frac{1}{2}(0.4055) = 0.2027$. The integrand decreases from $\frac{1}{3} \approx 0.333$ at $x = 2$ to $\frac{1}{8} = 0.125$ at $x = 3$ over an interval of length 1, and the midpoint value $\frac{1}{2.5^2 - 1} \approx 0.190$ is close to 0.2027, so the answer is reasonable.

Answer. $\frac{1}{2} \ln \frac{3}{2} \approx 0.2027$

16. Evaluate

$$\int \frac{3x^2 + 2}{x^2 + 1} dx.$$

SOLUTION

Step 1. Compare degrees: the numerator $3x^2 + 2$ and the denominator $x^2 + 1$ both have degree 2. Since the numerator degree is not smaller than the denominator degree, partial fractions cannot be applied directly; divide first.

Step 2. Long division: the leading terms give $\frac{3x^2}{x^2} = 3$. Multiply back: $3(x^2 + 1) = 3x^2 + 3$. Subtract: $(3x^2 + 2) - (3x^2 + 3) = -1$.

Step 3. The remainder -1 has degree $0 < 2$, so we stop. Thus $3x^2 + 2 = 3(x^2 + 1) - 1$.

Step 4. Divide by $x^2 + 1$:

$$\frac{3x^2 + 2}{x^2 + 1} = 3 - \frac{1}{x^2 + 1}.$$

Step 5. Check: $3 - \frac{1}{x^2 + 1} = \frac{3(x^2 + 1) - 1}{x^2 + 1} = \frac{3x^2 + 3 - 1}{x^2 + 1} = \frac{3x^2 + 2}{x^2 + 1}$.

Step 6. The remaining fraction already has an irreducible denominator and a constant numerator, so no further decomposition is needed.

Step 7. Integrate term by term: $\int 3 dx = 3x$ and, by the standard form $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right)$ with $a = 1$, $\int \frac{dx}{x^2 + 1} = \arctan x$.

Step 8. Result:

$$\int \frac{3x^2 + 2}{x^2 + 1} dx = 3x - \arctan x + C.$$

Step 9. Verify by differentiating: $\frac{d}{dx}[3x - \arctan x] = 3 - \frac{1}{1 + x^2} = \frac{3x^2 + 3 - 1}{x^2 + 1} = \frac{3x^2 + 2}{x^2 + 1}$.

Answer. $3x - \arctan x + C$

17. Evaluate

$$\int \frac{x + 1}{x^2(x - 2)} dx.$$

SOLUTION

Step 1. The numerator has degree 1 and the denominator degree 3, so no long division is needed. The denominator is already factored: $x^2 = (x - 0)^2$ is a repeated linear factor and $x - 2$ is a distinct linear factor.

Step 2. A repeated factor x^2 needs a term for every power from 1 to 2, and $x - 2$ needs one term:

$$\frac{x+1}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2}.$$

Step 3. Multiply by $x^2(x-2)$:

$$x+1 = Ax(x-2) + B(x-2) + Cx^2.$$

Step 4. Substitute $x = 0$: $1 = 0 + B(-2) + 0$, so $B = -\frac{1}{2}$.

Step 5. Substitute $x = 2$: $3 = 0 + 0 + C(4)$, so $C = \frac{3}{4}$.

Step 6. Expand the right side to compare coefficients: $Ax^2 - 2Ax + Bx - 2B + Cx^2 = (A+C)x^2 + (B-2A)x - 2B$.

Step 7. Match the x^2 coefficient (left side has 0): $A+C=0$, so $A=-C=-\frac{3}{4}$.

Step 8. Check with the x coefficient: $B-2A = -\frac{1}{2} + \frac{3}{2} = 1$, which matches the 1 on the left. Check the constant: $-2B = 1$, which matches. So

$$\frac{x+1}{x^2(x-2)} = -\frac{3/4}{x} - \frac{1/2}{x^2} + \frac{3/4}{x-2}.$$

Step 9. Integrate the logarithmic terms with $\int \frac{du}{u} = \ln|u|$: $-\frac{3}{4} \ln|x|$ and $\frac{3}{4} \ln|x-2|$.

Step 10. Integrate the repeated-factor term with the power rule: $-\frac{1}{2} \int x^{-2} dx = -\frac{1}{2} \cdot \frac{x^{-1}}{-1} = \frac{1}{2x}$.

Step 11. Combine:

$$\int \frac{x+1}{x^2(x-2)} dx = -\frac{3}{4} \ln|x| + \frac{1}{2x} + \frac{3}{4} \ln|x-2| + C.$$

Step 12. Verify at $x = 1$: the integrand is $\frac{2}{1 \cdot (-1)} = -2$. The derivative of the

answer is $-\frac{3}{4} \cdot \frac{1}{1} - \frac{1}{2 \cdot 1^2} + \frac{3}{4} \cdot \frac{1}{-1} = -\frac{3}{4} - \frac{1}{2} - \frac{3}{4} = -2$.

Answer. $-\frac{3}{4} \ln|x| + \frac{1}{2x} + \frac{3}{4} \ln|x-2| + C$

18. Evaluate

$$\int \frac{2x}{(x^2+1)(x-1)} dx.$$

SOLUTION

Step 1. The numerator has degree 1 and the denominator degree 3, so no long division is needed. The factor x^2+1 is irreducible over the reals and $x-1$ is a distinct linear factor.

Step 2. The irreducible quadratic needs a linear numerator and the linear factor a constant:

$$\frac{2x}{(x^2 + 1)(x - 1)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1}.$$

Step 3. Multiply by $(x^2 + 1)(x - 1)$:

$$2x = (Ax + B)(x - 1) + C(x^2 + 1).$$

Step 4. Substitute $x = 1$ (kills the first term): $2 = 0 + C(2)$, so $C = 1$.

Step 5. Expand the right side: $(Ax + B)(x - 1) = Ax^2 - Ax + Bx - B$, so

$$2x = (A + C)x^2 + (B - A)x + (C - B).$$

Step 6. Match the x^2 coefficient: $A + C = 0$, so $A = -1$.

Step 7. Match the constant term: $C - B = 0$, so $B = C = 1$.

Step 8. Check with the x coefficient: $B - A = 1 - (-1) = 2$, which matches the 2 on the left. So

$$\frac{2x}{(x^2 + 1)(x - 1)} = \frac{-x + 1}{x^2 + 1} + \frac{1}{x - 1}.$$

Step 9. Check at $x = 0$: left side 0; right side $\frac{1}{1} + \frac{1}{-1} = 0$. Good.

Step 10. Split the first fraction: $\frac{-x + 1}{x^2 + 1} = -\frac{x}{x^2 + 1} + \frac{1}{x^2 + 1}$. For $\int \frac{x}{x^2 + 1} dx$ let $u = x^2 + 1$, $du = 2x dx$, giving $\frac{1}{2} \ln(x^2 + 1)$. For $\int \frac{dx}{x^2 + 1}$ use the standard form $\arctan x$.

Step 11. The third term is $\int \frac{dx}{x - 1} = \ln |x - 1|$. Combine:

$$\int \frac{2x}{(x^2 + 1)(x - 1)} dx = -\frac{1}{2} \ln(x^2 + 1) + \arctan x + \ln |x - 1| + C.$$

Step 12. Verify by differentiating: $-\frac{x}{x^2 + 1} + \frac{1}{x^2 + 1} + \frac{1}{x - 1} = \frac{-x + 1}{x^2 + 1} + \frac{1}{x - 1}$, which is the decomposition just checked.

Answer. $-\frac{1}{2} \ln(x^2 + 1) + \arctan x + \ln |x - 1| + C$

19. Evaluate

$$\int \frac{x^4 + 3x^2 - x + 1}{(x - 1)(x^2 + 1)} dx.$$

SOLUTION

Step 1. Compare degrees: the numerator has degree 4; the denominator $(x - 1)(x^2 + 1)$ has degree 3. The numerator degree is not smaller, so we must long-divide first.

Step 2. Expand the denominator: $(x - 1)(x^2 + 1) = x^3 + x - x^2 - 1 = x^3 - x^2 + x - 1$.

Step 3. Divide $x^4 + 0x^3 + 3x^2 - x + 1$ by $x^3 - x^2 + x - 1$. Leading terms:

$\frac{x^4}{x^3} = x$. Multiply back: $x(x^3 - x^2 + x - 1) = x^4 - x^3 + x^2 - x$. Subtract: $(x^4 + 0x^3 + 3x^2 - x + 1) - (x^4 - x^3 + x^2 - x) = x^3 + 2x^2 + 0x + 1$.

Step 4. Next: $\frac{x^3}{x^3} = 1$. Multiply back: $1 \cdot (x^3 - x^2 + x - 1)$. Subtract: $(x^3 + 2x^2 + 0x + 1) - (x^3 - x^2 + x - 1) = 3x^2 - x + 2$.

Step 5. The remainder $3x^2 - x + 2$ has degree $2 < 3$, so the division stops. The quotient is $x + 1$, and

$$\frac{x^4 + 3x^2 - x + 1}{(x - 1)(x^2 + 1)} = x + 1 + \frac{3x^2 - x + 2}{(x - 1)(x^2 + 1)}.$$

Step 6. Decompose the proper fraction:

$$\frac{3x^2 - x + 2}{(x - 1)(x^2 + 1)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1}, \quad 3x^2 - x + 2 = (Ax + B)(x - 1) + C(x^2 + 1).$$

Step 7. Substitute $x = 1$: $3 - 1 + 2 = 4 = C(2)$, so $C = 2$.

Step 8. Expand the right side: $(A + C)x^2 + (B - A)x + (C - B)$. Match x^2 : $A + C = 3$, so $A = 1$. Match the constant: $C - B = 2$, so $B = 0$.

Step 9. Check with the x coefficient: $B - A = 0 - 1 = -1$, which matches the -1 on the left. So

$$\frac{3x^2 - x + 2}{(x - 1)(x^2 + 1)} = \frac{x}{x^2 + 1} + \frac{2}{x - 1}.$$

Step 10. The full integrand is therefore $x + 1 + \frac{x}{x^2 + 1} + \frac{2}{x - 1}$.

Step 11. Integrate: $\int (x + 1) dx = \frac{x^2}{2} + x$. For $\int \frac{x}{x^2 + 1} dx$ let $u = x^2 + 1$, $du = 2x dx$, giving $\frac{1}{2} \ln(x^2 + 1)$. For $\int \frac{2}{x - 1} dx = 2 \ln |x - 1|$.

Step 12. Combine:

$$\int \frac{x^4 + 3x^2 - x + 1}{(x - 1)(x^2 + 1)} dx = \frac{x^2}{2} + x + \frac{1}{2} \ln(x^2 + 1) + 2 \ln |x - 1| + C.$$

Step 13. Verify at $x = 2$: the integrand is $\frac{16 + 12 - 2 + 1}{(1)(5)} = \frac{27}{5} = 5.4$. The

derivative of the answer is $2 + 1 + \frac{2}{5} + \frac{2}{1} = 5.4$.

Answer. $\frac{x^2}{2} + x + \frac{1}{2} \ln(x^2 + 1) + 2 \ln |x - 1| + C$

The Weierstrass Substitution

20. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int \frac{1}{1 + \sin x} dx.$$

SOLUTION

Step 1. Use the Weierstrass substitution $t = \tan\left(\frac{x}{2}\right)$ with $\sin x = \frac{2t}{1+t^2}$ and $dx = \frac{2 dt}{1+t^2}$.

Step 2. Rewrite the denominator over the common denominator $1+t^2$:

$$1 + \sin x = 1 + \frac{2t}{1+t^2} = \frac{1+t^2+2t}{1+t^2} = \frac{(1+t)^2}{1+t^2},$$

using the perfect square $t^2 + 2t + 1 = (t+1)^2$.

Step 3. Substitute and take the reciprocal:

$$\int \frac{1+t^2}{(1+t)^2} \cdot \frac{2 dt}{1+t^2}.$$

Step 4. Cancel $1+t^2$:

$$\int \frac{2 dt}{(1+t)^2}.$$

Step 5. This needs no partial fractions. Let $u = 1+t$, $du = dt$: $2 \int u^{-2} du$.

Step 6. By the power rule $\int u^{-2} du = \frac{u^{-1}}{-1} = -\frac{1}{u}$, so the integral is $-\frac{2}{u} = -\frac{2}{1+t}$.

Step 7. Substitute back $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{1}{1+\sin x} dx = -\frac{2}{1+\tan(x/2)} + C.$$

Step 8. Verify by differentiating, using $\frac{dt}{dx} = \frac{1+t^2}{2}$: $\frac{d}{dx} \left[-\frac{2}{1+t} \right] = \frac{2}{(1+t)^2} \cdot$

$\frac{1+t^2}{2} = \frac{1+t^2}{(1+t)^2}$, which equals $\frac{1}{1+\sin x}$ by the second step.

Answer. $-\frac{2}{1+\tan(x/2)} + C$

21. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int_0^{\pi/2} \frac{1}{1+\sin x + \cos x} dx.$$

SOLUTION

Step 1. The denominator mixes $\sin x$ and $\cos x$, so use the Weierstrass substitution

$t = \tan\left(\frac{x}{2}\right)$, with $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2 dt}{1+t^2}$.

Step 2. Change the limits. When $x = 0$: $t = \tan 0 = 0$. When $x = \frac{\pi}{2}$: $t = \tan \frac{\pi}{4} =$

1. The function $\tan\left(\frac{x}{2}\right)$ is continuous on $\left[0, \frac{\pi}{2}\right]$, so the substitution is valid on the whole interval.

Step 3. Rewrite the denominator over $1 + t^2$:

$$1 + \sin x + \cos x = \frac{(1 + t^2) + 2t + (1 - t^2)}{1 + t^2} = \frac{2 + 2t}{1 + t^2} = \frac{2(1 + t)}{1 + t^2}.$$

Step 4. Substitute and take the reciprocal:

$$\int_0^1 \frac{1 + t^2}{2(1 + t)} \cdot \frac{2 dt}{1 + t^2}.$$

Step 5. Cancel the factor $1 + t^2$ and the factor 2:

$$\int_0^1 \frac{dt}{1 + t}.$$

Step 6. Integrate with $u = 1 + t$, $du = dt$: $\int \frac{dt}{1 + t} = \ln |1 + t|$.

Step 7. Evaluate between the new limits: $\ln(1 + 1) - \ln(1 + 0) = \ln 2 - \ln 1 = \ln 2 - 0 = \ln 2$.

Step 8. Numerical check: $\ln 2 \approx 0.6931$. At $x = 0$ the integrand is $\frac{1}{0 + 1} = 1$, at

$x = \frac{\pi}{2}$ it is also $\frac{1}{1 + 0} = 1$, and at $x = \frac{\pi}{4}$ it is $\frac{1}{1 + \sqrt{2}} \approx 0.414$. The average is near 0.44, and $0.44 \times \frac{\pi}{2} \approx 0.69$, consistent with $\ln 2$.

Answer. $\ln 2 \approx 0.6931$

22. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int \frac{1}{3 + 5 \cos x} dx,$$

and simplify the resulting expression fully.

SOLUTION

Step 1. Use $t = \tan\left(\frac{x}{2}\right)$ with $\cos x = \frac{1 - t^2}{1 + t^2}$ and $dx = \frac{2 dt}{1 + t^2}$.

Step 2. Rewrite the denominator over $1 + t^2$:

$$3 + 5 \cos x = \frac{3(1 + t^2) + 5(1 - t^2)}{1 + t^2} = \frac{3 + 3t^2 + 5 - 5t^2}{1 + t^2} = \frac{8 - 2t^2}{1 + t^2}.$$

Step 3. Substitute and take the reciprocal:

$$\int \frac{1 + t^2}{8 - 2t^2} \cdot \frac{2 dt}{1 + t^2} = \int \frac{2 dt}{8 - 2t^2}.$$

Step 4. Factor 2 out of the denominator and cancel: $\frac{2}{8 - 2t^2} = \frac{2}{2(4 - t^2)} = \frac{1}{4 - t^2}$, so the integral is

$$\int \frac{dt}{4-t^2}.$$

Step 5. Decide between arctangent and logarithm: $4-t^2$ is a difference of squares and factors over the reals, $4-t^2 = (2-t)(2+t)$, so partial fractions will give logarithms.

Step 6. Set up $\frac{1}{(2-t)(2+t)} = \frac{A}{2-t} + \frac{B}{2+t}$, so $1 = A(2+t) + B(2-t)$.

Step 7. Substitute $t = 2$: $1 = A(4)$, so $A = \frac{1}{4}$. Substitute $t = -2$: $1 = B(4)$, so $B = \frac{1}{4}$. Check: $\frac{1}{4}(2+t) + \frac{1}{4}(2-t) = \frac{4}{4} = 1$.

Step 8. Integrate: $\int \frac{dt}{2+t} = \ln|2+t|$ and $\int \frac{dt}{2-t} = -\ln|2-t|$ (let $u = 2-t$, $du = -dt$). Hence

$$\int \frac{dt}{4-t^2} = \frac{1}{4} \ln|2+t| - \frac{1}{4} \ln|2-t| = \frac{1}{4} \ln \left| \frac{2+t}{2-t} \right|.$$

Step 9. Substitute back $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{dx}{3+5\cos x} = \frac{1}{4} \ln \left| \frac{2+\tan(x/2)}{2-\tan(x/2)} \right| + C.$$

Step 10. Verify by differentiating, using $\frac{dt}{dx} = \frac{1+t^2}{2}$: $\frac{1}{4} \left(\frac{1}{2+t} + \frac{1}{2-t} \right) \frac{1+t^2}{2} = \frac{1}{4} \cdot \frac{4}{4-t^2} \cdot \frac{1+t^2}{2} = \frac{1+t^2}{2(4-t^2)} = \frac{1+t^2}{8-2t^2} = \frac{1}{3+5\cos x}$ by the second step.

Answer. $\frac{1}{4} \ln \left| \frac{2+\tan(x/2)}{2-\tan(x/2)} \right| + C$

23. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int \frac{1}{2+\sin x} dx.$$

SOLUTION

Step 1. Use $t = \tan\left(\frac{x}{2}\right)$ with $\sin x = \frac{2t}{1+t^2}$ and $dx = \frac{2 dt}{1+t^2}$.

Step 2. Rewrite the denominator over $1+t^2$:

$$2+\sin x = \frac{2(1+t^2)+2t}{1+t^2} = \frac{2t^2+2t+2}{1+t^2}.$$

Step 3. Substitute and take the reciprocal:

$$\int \frac{1+t^2}{2t^2+2t+2} \cdot \frac{2 dt}{1+t^2} = \int \frac{2 dt}{2t^2+2t+2} = \int \frac{dt}{t^2+t+1}.$$

Step 4. Check the discriminant of t^2+t+1 : $b^2-4ac = 1-4 = -3 < 0$, so it

does not factor over the reals and the result will be an arctangent.

Step 5. Complete the square: $t^2 + t + 1 = \left(t + \frac{1}{2}\right)^2 - \frac{1}{4} + 1 = \left(t + \frac{1}{2}\right)^2 + \frac{3}{4}$.

Step 6. Let $u = t + \frac{1}{2}$, $du = dt$, and $a = \frac{\sqrt{3}}{2}$ (since $a^2 = \frac{3}{4}$). Use $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right)$:

$$\int \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} = \frac{2}{\sqrt{3}} \arctan\left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right).$$

Step 7. Simplify the argument by multiplying top and bottom by 2: $\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{2t + 1}{\sqrt{3}}$.

Step 8. Substitute back $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{dx}{2 + \sin x} = \frac{2}{\sqrt{3}} \arctan\left(\frac{2 \tan(x/2) + 1}{\sqrt{3}}\right) + C.$$

Step 9. Verify by differentiating, using $\frac{dt}{dx} = \frac{1+t^2}{2}$: $\frac{2}{\sqrt{3}} \cdot \frac{1}{1 + \frac{(2t+1)^2}{3}} \cdot \frac{2}{\sqrt{3}} \cdot \frac{1+t^2}{2} = \frac{4}{3} \cdot \frac{3}{3 + (2t+1)^2} \cdot \frac{1+t^2}{2} = \frac{2(1+t^2)}{4t^2 + 4t + 4} = \frac{1+t^2}{2t^2 + 2t + 2}$, which equals $\frac{1}{2 + \sin x}$ by the second step.

Answer. $\frac{2}{\sqrt{3}} \arctan\left(\frac{2 \tan(x/2) + 1}{\sqrt{3}}\right) + C$

24. Show that $\int \sec x \, dx$ can be evaluated with the substitution $t = \tan\left(\frac{x}{2}\right)$, and confirm that the result agrees with the memorised formula $\ln |\sec x + \tan x| + C$.

SOLUTION

Step 1. Write the integrand as a rational function of $\cos x$: $\sec x = \frac{1}{\cos x}$. Use

$t = \tan\left(\frac{x}{2}\right)$ with $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2 \, dt}{1+t^2}$.

Step 2. Then $\sec x = \frac{1+t^2}{1-t^2}$, and

$$\int \sec x \, dx = \int \frac{1+t^2}{1-t^2} \cdot \frac{2 \, dt}{1+t^2} = \int \frac{2 \, dt}{1-t^2}.$$

Step 3. The denominator is a difference of squares, $1-t^2 = (1-t)(1+t)$, so use partial fractions: $\frac{2}{(1-t)(1+t)} = \frac{A}{1-t} + \frac{B}{1+t}$, giving $2 = A(1+t) + B(1-t)$.

Step 4. Substitute $t = 1$: $2 = A(2)$, so $A = 1$. Substitute $t = -1$: $2 = B(2)$, so

$B = 1$. Check: $(1 + t) + (1 - t) = 2$. So

$$\frac{2}{1 - t^2} = \frac{1}{1 - t} + \frac{1}{1 + t}.$$

Step 5. Integrate: $\int \frac{dt}{1 + t} = \ln |1 + t|$ and $\int \frac{dt}{1 - t} = -\ln |1 - t|$ (let $u = 1 - t$, $du = -dt$). Therefore

$$\int \sec x \, dx = \ln |1 + t| - \ln |1 - t| + C = \ln \left| \frac{1 + t}{1 - t} \right| + C = \ln \left| \frac{1 + \tan(x/2)}{1 - \tan(x/2)} \right| + C.$$

Step 6. Now show this equals $\ln |\sec x + \tan x|$. Write $c = \cos\left(\frac{x}{2}\right)$ and $s = \sin\left(\frac{x}{2}\right)$, so $t = \frac{s}{c}$. Then

$$\frac{1 + t}{1 - t} = \frac{1 + \frac{s}{c}}{1 - \frac{s}{c}} = \frac{\frac{c+s}{c}}{\frac{c-s}{c}} = \frac{c + s}{c - s}.$$

Step 7. Multiply numerator and denominator by $c + s$ (allowed whenever $c + s \neq 0$):

$$\frac{c + s}{c - s} = \frac{(c + s)^2}{(c - s)(c + s)}.$$

Step 8. Expand the numerator: $(c + s)^2 = c^2 + 2cs + s^2 = (c^2 + s^2) + 2cs$. Use $c^2 + s^2 = \cos^2\left(\frac{x}{2}\right) + \sin^2\left(\frac{x}{2}\right) = 1$ (Pythagorean identity) and $2cs = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) = \sin x$ (double-angle identity). So the numerator is $1 + \sin x$.

Step 9. Expand the denominator with the difference of squares: $(c - s)(c + s) = c^2 - s^2 = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right) = \cos x$ (double-angle identity).

Step 10. Therefore

$$\frac{1 + t}{1 - t} = \frac{1 + \sin x}{\cos x} = \frac{1}{\cos x} + \frac{\sin x}{\cos x} = \sec x + \tan x.$$

Step 11. This is an exact equality of the expressions inside the logarithm, so $\ln \left| \frac{1 + \tan(x/2)}{1 - \tan(x/2)} \right| = \ln |\sec x + \tan x|$ with no extra constant. The Weierstrass result and the memorised formula are the same antiderivative:

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C.$$

Step 12. Numerical check at $x = 0$: $t = 0$, so $\frac{1 + 0}{1 - 0} = 1$, and $\sec 0 + \tan 0 = 1 + 0 = 1$. At $x = \frac{\pi}{2}$ both expressions are undefined, consistent with $\sec x$ having a vertical asymptote there.

Answer. $\ln \left| \frac{1 + \tan(x/2)}{1 - \tan(x/2)} \right| + C = \ln |\sec x + \tan x| + C$, because $\frac{1 + \tan(x/2)}{1 - \tan(x/2)} = \frac{(c + s)^2}{c^2 - s^2} = \frac{1 + \sin x}{\cos x} = \sec x + \tan x$

Mixed Technique Identification

25. For each integral, state which single technique from Weeks 3 and 4 (substitution, integration by parts, trig-power identities, partial fractions, or the Weierstrass substitution) is the most efficient, explain why, and then evaluate it: (a) $\int \frac{x}{x^2 - 9} dx$; (b) $\int x^2 \cos(x^3) dx$; (c) $\int \frac{1}{\sin x - \cos x} dx$; (d) $\int \sec^6 x dx$.

SOLUTION

Step 1. Part (a), choice: SUBSTITUTION with $u = x^2 - 9$. The numerator x is, up to the constant factor $\frac{1}{2}$, the derivative of the denominator $x^2 - 9$ (since $\frac{d}{dx}(x^2 - 9) = 2x$), which is the pattern $\int \frac{u'}{u}$. Partial fractions would also work after factoring $x^2 - 9 = (x - 3)(x + 3)$ but takes more steps; the Weierstrass substitution is irrelevant because there is no trigonometric function.

Step 2. Part (a), evaluation: let $u = x^2 - 9$, so $du = 2x dx$ and $x dx = \frac{du}{2}$. Then

$$\int \frac{x}{x^2 - 9} dx = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |x^2 - 9| + C.$$

Verify: $\frac{d}{dx} \left[\frac{1}{2} \ln |x^2 - 9| \right] = \frac{1}{2} \cdot \frac{2x}{x^2 - 9} = \frac{x}{x^2 - 9}.$

Step 3. Part (b), choice: SUBSTITUTION with $u = x^3$. The inner function of the cosine is x^3 and its derivative is $3x^2$, and the factor x^2 in front of the cosine is (up to the constant 3) exactly that derivative. Integration by parts would make the integral harder, and no trigonometric-power identity helps because the argument is x^3 , not x .

Step 4. Part (b), evaluation: let $u = x^3$, so $du = 3x^2 dx$ and $x^2 dx = \frac{du}{3}$. Then

$$\int x^2 \cos(x^3) dx = \frac{1}{3} \int \cos u du = \frac{1}{3} \sin u + C = \frac{1}{3} \sin(x^3) + C.$$

Verify: $\frac{d}{dx} \left[\frac{1}{3} \sin(x^3) \right] = \frac{1}{3} \cos(x^3) \cdot 3x^2 = x^2 \cos(x^3).$

Step 5. Part (c), choice: WEIERSTRASS SUBSTITUTION $t = \tan\left(\frac{x}{2}\right)$. The integrand is a rational function of $\sin x$ and $\cos x$ with a sum $\sin x - \cos x$ in the denominator. There is no power of sine or cosine to peel off, no secant or tangent structure, and no u whose derivative appears in the numerator (the numerator is just 1), so the odd-power, even-power and sec/tan rules and plain substitution all fail. Partial fractions alone does not apply because the integrand is not yet rational in x . The Weierstrass substitution always works for such integrands.

Step 6. Part (c), evaluation, step 1: use $\sin x = \frac{2t}{1 + t^2}$, $\cos x = \frac{1 - t^2}{1 + t^2}$, $dx = \frac{2 dt}{1 + t^2}$. Then

$$\sin x - \cos x = \frac{2t - (1 - t^2)}{1 + t^2} = \frac{t^2 + 2t - 1}{1 + t^2}.$$

Step 7. Part (c), evaluation, step 2: substitute and take the reciprocal:

$$\int \frac{1+t^2}{t^2+2t-1} \cdot \frac{2dt}{1+t^2} = \int \frac{2dt}{t^2+2t-1}.$$

Step 8. Part (c), evaluation, step 3: the discriminant of t^2+2t-1 is $4+4=8>0$, so it factors over the reals and the result is a logarithm. Complete the square: $t^2+2t-1=(t+1)^2-2$. Let $u=t+1$, $a=\sqrt{2}$.

Step 9. Part (c), evaluation, step 4: use $\int \frac{du}{u^2-a^2} = \frac{1}{2a} \ln \left| \frac{u-a}{u+a} \right|$ (from partial fractions $\frac{1}{u^2-a^2} = \frac{1}{2a} \left(\frac{1}{u-a} - \frac{1}{u+a} \right)$). With $a=\sqrt{2}$, $2 \cdot \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$, so

$$\int \frac{1}{\sin x - \cos x} dx = \frac{1}{\sqrt{2}} \ln \left| \frac{t+1-\sqrt{2}}{t+1+\sqrt{2}} \right| + C = \frac{1}{\sqrt{2}} \ln \left| \frac{\tan(x/2)+1-\sqrt{2}}{\tan(x/2)+1+\sqrt{2}} \right| + C.$$

Step 10. Part (c), verification: differentiating with $\frac{dt}{dx} = \frac{1+t^2}{2}$ gives

$$\frac{1}{\sqrt{2}} \left(\frac{1}{t+1-\sqrt{2}} - \frac{1}{t+1+\sqrt{2}} \right) \frac{1+t^2}{2} = \frac{1}{\sqrt{2}} \cdot \frac{2\sqrt{2}}{(t+1)^2-2} \cdot \frac{1+t^2}{2} = \frac{1+t^2}{t^2+2t-1} = \frac{1}{\sin x - \cos x}.$$

Step 11. Part (d), choice: the SEC/TAN even-secant rule (a trig-power technique) with $u = \tan x$. The power of secant is 6, which is even, so one factor $\sec^2 x = \frac{d}{dx} \tan x$ can be peeled off, and the remaining $\sec^4 x$ is converted to tangents by $\sec^2 x = 1 + \tan^2 x$. Using the Weierstrass substitution here would produce a rational function of degree 10 in t , far messier for no benefit.

Step 12. Part (d), evaluation, step 1: write $\sec^6 x = \sec^4 x \cdot \sec^2 x = (\sec^2 x)^2 \sec^2 x = (1 + \tan^2 x)^2 \sec^2 x$.

Step 13. Part (d), evaluation, step 2: let $u = \tan x$, $du = \sec^2 x dx$. Expand $(1+u^2)^2 = 1+2u^2+u^4$. Then

$$\int \sec^6 x dx = \int (1+2u^2+u^4) du = u + \frac{2u^3}{3} + \frac{u^5}{5} + C.$$

Step 14. Part (d), evaluation, step 3: substitute back:

$$\int \sec^6 x dx = \tan x + \frac{2}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C.$$

Verify: the derivative is $\sec^2 x + 2 \tan^2 x \sec^2 x + \tan^4 x \sec^2 x = \sec^2 x (1 + 2 \tan^2 x + \tan^4 x) = \sec^2 x (1 + \tan^2 x)^2 = \sec^2 x \cdot \sec^4 x = \sec^6 x$.

Answer. (a) substitution: $\frac{1}{2} \ln |x^2 - 9| + C$; (b) substitution: $\frac{1}{3} \sin(x^3) + C$; (c)

Weierstrass substitution: $\frac{1}{\sqrt{2}} \ln \left| \frac{\tan(x/2)+1-\sqrt{2}}{\tan(x/2)+1+\sqrt{2}} \right| + C$; (d) even-secant rule with

$$u = \tan x: \tan x + \frac{2}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C$$

26. For each integral, name the most efficient single technique (substitution, integration by parts, trig-power identities, partial fractions, or the Weierstrass substitution) and then evaluate it: (a) $\int x \cos x \, dx$; (b) $\int \frac{1}{x^2 - 4x + 3} \, dx$; (c) $\int \cos^3 x \, dx$.

SOLUTION

Step 1. Part (a), choice: INTEGRATION BY PARTS. The integrand is a product of a polynomial x and a trigonometric function $\cos x$, and differentiating the polynomial factor simplifies it (its derivative is 1), which is the situation for which $\int u \, dv = uv - \int v \, du$ is designed. Substitution fails because $\cos x$ is not the derivative of any inner function appearing in x .

Step 2. Part (a), evaluation: choose $u = x$ and $dv = \cos x \, dx$. Then $du = dx$ and $v = \sin x$. Apply the formula:

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx = x \sin x - (-\cos x) + C = x \sin x + \cos x + C.$$

Verify: $\frac{d}{dx}[x \sin x + \cos x] = \sin x + x \cos x - \sin x = x \cos x$.

Step 3. Part (b), choice: PARTIAL FRACTIONS. The integrand is a rational function whose denominator is a quadratic that factors over the reals, and the numerator is a constant, with no derivative-of-the-denominator pattern for substitution to use. Factor the quadratic: $x^2 - 4x + 3 = (x - 1)(x - 3)$, since $(-1) + (-3) = -4$ and $(-1)(-3) = 3$.

Step 4. Part (b), evaluation, step 1: set up $\frac{1}{(x - 1)(x - 3)} = \frac{A}{x - 1} + \frac{B}{x - 3}$, so $1 = A(x - 3) + B(x - 1)$. Substitute $x = 1$: $1 = A(-2)$, so $A = -\frac{1}{2}$. Substitute $x = 3$: $1 = B(2)$, so $B = \frac{1}{2}$. Check: $-\frac{1}{2}(x - 3) + \frac{1}{2}(x - 1) = \frac{-x + 3 + x - 1}{2} = 1$.

Step 5. Part (b), evaluation, step 2: integrate with $\int \frac{du}{u} = \ln |u|$:

$$\int \frac{dx}{x^2 - 4x + 3} = -\frac{1}{2} \ln |x - 1| + \frac{1}{2} \ln |x - 3| + C = \frac{1}{2} \ln \left| \frac{x - 3}{x - 1} \right| + C.$$

Verify: $-\frac{1}{2(x - 1)} + \frac{1}{2(x - 3)} = \frac{-(x - 3) + (x - 1)}{2(x - 1)(x - 3)} = \frac{2}{2(x - 1)(x - 3)} = \frac{1}{x^2 - 4x + 3}$.

Step 6. Part (c), choice: TRIG-POWER IDENTITIES (odd power of cosine). The power of cosine is 3, which is odd, so one factor $\cos x$ can be peeled off and the remaining even power $\cos^2 x$ converted using $\cos^2 x = 1 - \sin^2 x$, after which $u = \sin x$ works. Substitution alone fails because $\cos^3 x \, dx$ is not du for any simple u until the factor is split.

Step 7. Part (c), evaluation, step 1: write $\cos^3 x = \cos^2 x \cdot \cos x = (1 - \sin^2 x) \cos x$ using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$.

Step 8. Part (c), evaluation, step 2: let $u = \sin x$, so $du = \cos x \, dx$. Then

$$\int \cos^3 x \, dx = \int (1 - u^2) \, du = u - \frac{u^3}{3} + C = \sin x - \frac{\sin^3 x}{3} + C.$$

Verify: $\frac{d}{dx} \left[\sin x - \frac{\sin^3 x}{3} \right] = \cos x - \sin^2 x \cos x = \cos x (1 - \sin^2 x) = \cos^3 x$.

Answer. (a) integration by parts: $x \sin x + \cos x + C$; (b) partial fractions: $\frac{1}{2} \ln \left| \frac{x-3}{x-1} \right| + C$; (c) odd power of cosine with $u = \sin x$: $\sin x - \frac{\sin^3 x}{3} + C$